

DLPC3479 显示和光控制器

1 特性

- 适用于 DLP4710 (.47 1080p) TRP DMD 的 2xDLPC3479 显示和光控制器
- 显示 功能:
 - 最高支持 1080p 的输入图像大小
 - 输入帧速率可达 60Hz (2D 和 3D)
 - 24 位输入像素接口支持
 - 并行或 BT656 接口协议
 - 高达 150MHz 的像素时钟
 - 图像处理: IntelliBright™ 算法、图像大小调整、色彩坐标调整 (CCA)、可编程 Degamma
- 光控制 功能:
 - 针对机器视觉和数码曝光进行了优化的图形显示
 - 灵活的内部 (1D) 和外部 (2D) 图形流模式
 - 可编程曝光时间
 - 高达 1440Hz (1 位) 和 180Hz (8 位) 的高速图形速率
 - 可编程 2D 静态图形 (通过启动界面)
 - 内部图形流模式可简化系统设计
 - 无需视频接口
 - 在闪存中存储超过 1000 个图形
 - 用于实现摄像头或传感器同步的灵活触发器信号
 - 一个可配置输入触发器
 - 两个可配置输入触发器
- 系统 功能:
 - 器件配置的 I²C 控制

- 可编程启动界面
- 可编程 LED 电流控制
- 兼容 DLPA3000 和 DLPA3005 PMIC/LED 驱动器

2 应用

- 移动式附件全高清投影仪
- 低延迟游戏和可穿戴显示屏
- 3D 深度捕捉: 3D 照相机、3D 重建、AR/VR、牙科扫描仪
- 3D 机器视觉: 机器人学、计量学、直列式检测 (AOI)
- 3D 生物特征识别: 人脸和指纹识别
- 曝光: 3D 打印机、激光打标

3 说明

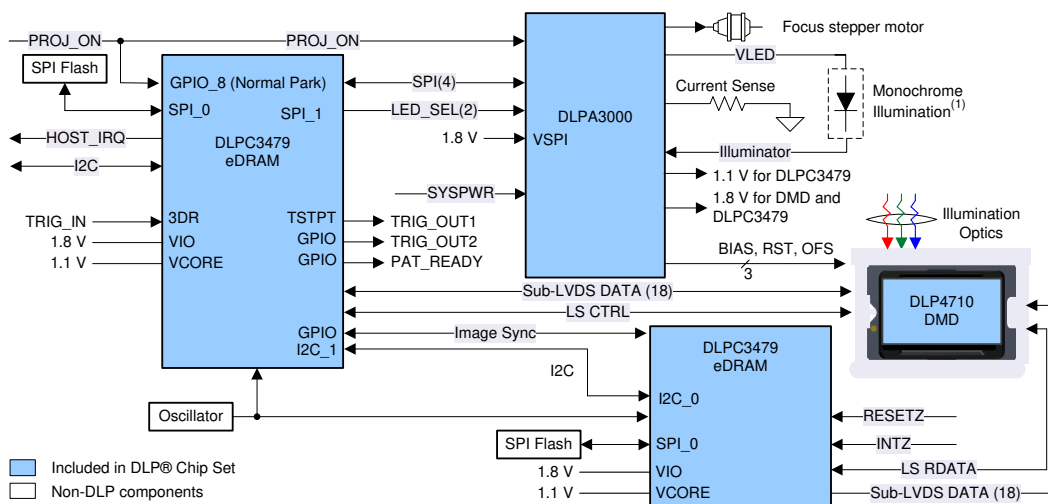
2xDLPC3479 显示和光控制器可支持视频显示和光控制应用中的 DLP4710 数字微镜器件 (DMD) 实现可靠运行。DLPC3479 控制器能够方便地将用户电子产品和 DMD 连接, 以高速、精确且高效地显示视频和控制光图形。

器件信息⁽¹⁾

器件型号	封装	封装尺寸 (标称值)
DLPC3479	NFBGA (201)	13.00mm x 13.00mm

(1) 如需了解所有可用封装, 请参阅数据表末尾的可订购产品附录。

简化原理图



目录

1	特性	1	8.3	Feature Description	31
2	应用	1	8.4	Device Functional Modes	47
3	说明	1	9	Application and Implementation	48
4	修订历史记录	2	9.1	Application Information	48
5	Pin Configuration and Functions	3	9.2	Typical Application	48
6	Specifications	14	10	Power Supply Recommendations	50
6.1	Absolute Maximum Ratings	14	10.1	DLPC3479 System Design Consideration	50
6.2	ESD Ratings	14	10.2	System Power-Up and Power-Down Sequence ...	50
6.3	Recommended Operating Conditions	15	10.3	DLPC3479 Power-Up Initialization Sequence	53
6.4	Thermal Information	15	10.4	DMD Fast PARK Control (PARKZ)	53
6.5	Electrical Characteristics over Recommended Operating Conditions	16	10.5	Hot Plug Usage	53
6.6	Electrical Characteristics	17	10.6	Maximum Signal Transition Time	53
6.7	High-Speed Sub-LVDS Electrical Characteristics ...	19	11	Layout	54
6.8	Low-Speed SDR Electrical Characteristics	20	11.1	Layout Guidelines	54
6.9	System Oscillators Timing Requirements	21	11.2	Layout Example	58
6.10	Power-Up and Reset Timing Requirements	21	11.3	Thermal Considerations	58
6.11	Parallel Interface Frame Timing Requirements ...	22	12	器件和文档支持	60
6.12	Parallel Interface General Timing Requirements ..	23	12.1	器件支持	60
6.13	Flash Interface Timing Requirements	24	12.2	文档支持	62
7	Parameter Measurement Information	28	12.3	接收文档更新通知	62
7.1	HOST_IRQ Usage Model	28	12.4	社区资源	62
7.2	Input Source	29	12.5	商标	62
8	Detailed Description	31	12.6	静电放电警告	62
8.1	Overview	31	12.7	Glossary	62
8.2	Functional Block Diagram	31	13	机械、封装和可订购信息	62

4 修订历史记录

Changes from Revision A (February 2019) to Revision B Page

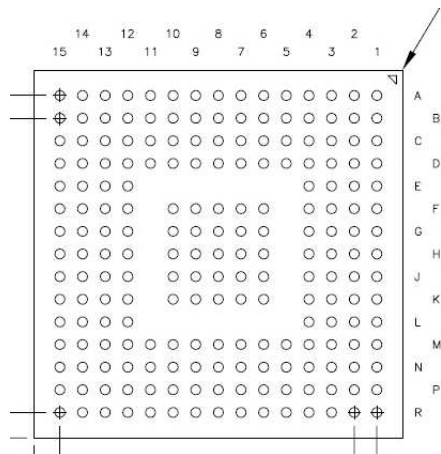
• Changed normal park time from 500 μ s to 20 ms	5
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Changes from Original (June 2018) to Revision A Page

• 已更改 将文档状态从预告信息更改为了生产数据	1
• Clarified pin description for TSTPT_2, TSTPT_4, STPT_5, and 3DR in <i>Pin Functions – Board Level Test, Debug, and Initialization</i> table	6
• Clarified pn descriptions for GPIO_07, GPIO_06, and GPIO_05 in <i>Pin Functions – GPIO Peripheral Interface</i> table	10
• Updated Pattern Display section	31
• Updated Content-Adaptive Illumination Control section	41

5 Pin Configuration and Functions

**ZEZ Package
201-Pin NFBGA
Bottom View**



	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
A	DMD_LS_C LK	DMD_LS_W DATA	DMD_HS_W DATAH_P	DMD_HS_W DATAG_P	DMD_HS_W DATAF_P	DMD_HS_W DATAE_P	DMD_HS_CLK P	DMD_HS_W DATAD_P	DMD_HS_W DATAC_P	DMD_HS_W DATAB_P	DMD_HS_W DATAA_P	CMP_OUT	SPI0_CLK	SPI0_CSZ0	CMP_PWM
B	DMD_DEN_ ARSTZ	DMD_LS_R DATA	DMD_HS_W DATAH_N	DMD_HS_W DATAG_N	DMD_HS_W DATAF_N	DMD_HS_W DATAE_N	DMD_HS_CLK N	DMD_HS_W DATAD_N	DMD_HS_W DATAC_N	DMD_HS_W DATAB_N	DMD_HS_W DATAA_N	SPI0_DIN	SPI0_DOUT	LED_SEL_1	LED_SEL_0
C	DD3P	DD3N	VDDL12	VSS	VDD	VSS	VCC	VSS	VCC	HWTEST_E N	RESETZ	SPI0_CSZ1	PARKZ	GPIO_00	GPIO_01
D	DD2P	DD2N	VDD	VCC	VDD	VSS	VDD	VSS	VDD	VSS	VCC_FLSH	VDD	VDD	GPIO_02	GPIO_03
E	DCLKP	DCLKN	VDD	VSS								VCC	VSS	GPIO_04	GPIO_05
F	DD1P	DD1N	RREF	VSS		VSS	VSS	VSS	VSS	VSS		VCC	VDD	GPIO_06	GPIO_07
G	DD0P	DD0N	VSS_PLLM	VSS		VSS	VSS	VSS	VSS	VSS		VSS	VSS	GPIO_08	GPIO_09
H	PLL_REFCL K_I	VDD_PLLM	VSS_PLLD	VSS		VSS	VSS	VSS	VSS	VSS		VSS	VDD	GPIO_10	GPIO_11
J	PLL_REFCL K_O	VDD_PLLD	VSS	VDD		VSS	VSS	VSS	VSS	VSS		VDD	VSS	GPIO_12	GPIO_13
K	PDATA_1	PDATA_0	VDD	VSS		VSS	VSS	VSS	VSS	VSS		VSS	VCC	GPIO_14	GPIO_15
L	PDATA_3	PDATA_2	VSS	VDD								VDD	VDD	GPIO_16	GPIO_17
M	PDATA_5	PDATA_4	VCC_INTF	VSS	VSS	VDD	VCC_INTF	VSS	VDD	VDD	VCC	VSS	JTAGTMS1	GPIO_18	GPIO_19
N	PDATA_7	PDATA_6	VCC_INTF	PDM_CVS_ TE	HSYNC_CS	3DR	VCC_INTF	HOST_IRQ	IIC0_SDA	IIC0_SCL	JTAGTMS2	JTAGTDO2	JTAGTDO1	TSTPT_6	TSTPT_7
P	VSYNCR_WE	DATEN_CM D	PCLK	PDATA_11	PDATA_13	PDATA_15	PDATA_17	PDATA_19	PDATA_21	PDATA_23	JTAGTRSTZ	JTAGTCK	JTAGTDI	TSTPT_4	TSTPT_5
R	PDATA_8	PDATA_9	PDATA_10	PDATA_12	PDATA_14	PDATA_16	PDATA_18	PDATA_20	PDATA_22	IIC1_SDA	IIC1_SCL	TSTPT_0	TSTPT_1	TSTPT_2	TSTPT_3

Figure 1. 13 mm × 13 mm Package – VF Ball Grid Array

- (1) If operation does not call for an external pullup and there is no external logic that might overcome the weak internal pulldown resistor, then this I/O can be left open or unconnected for normal operation. If operation does not call for an external pullup, but there is external logic that might overcome the weak internal pulldown resistor, then an external pulldown resistor is recommended to ensure a logic low.
- (2) External pullup resistor must be 8 k Ω , or less, for pins with internal pullup or down resistors.
- (3) If operation does not call for an external pullup and there is no external logic that might overcome the weak internal pulldown resistor, then the TSPT I/O can be left open/unconnected for normal operation. If operation does not call for an external pullup but there is external logic that might overcome the weak internal pulldown resistor, then an external pulldown resistor is recommended to ensure a logic low.

[illegible]

Pin Functions – Parallel Port Input Data and Control⁽¹⁾⁽²⁾

PIN		I/O	DESCRIPTION
NAME	NUMBER		PARALLEL RGB MODE
PCLK	P3	I ₁₁	Pixel clock ⁽³⁾
PDM_CVS_TE	N4	B ₅	Parallel data mask ⁽⁴⁾
VSNC_WE	P1	I ₁₁	Vsync ⁽⁵⁾
HSNC_CS	N5	I ₁₁	Hsync ⁽⁵⁾
DATAEN_CMD	P2	I ₁₁	Data Valid ⁽⁵⁾
PDATA_0 PDATA_1 PDATA_2 PDATA_3 PDATA_4 PDATA_5 PDATA_6 PDATA_7	K2 K1 L2 L1 M2 M1 N2 N1	I ₁₁	(TYPICAL RGB 888) Blue (bit weight 1) Blue (bit weight 2) Blue (bit weight 4) Blue (bit weight 8) Blue (bit weight 16) Blue (bit weight 32) Blue (bit weight 64) Blue (bit weight 128)
PDATA_8 PDATA_9 PDATA_10 PDATA_11 PDATA_12 PDATA_13 PDATA_14 PDATA_15	R1 R2 R3 P4 R4 P5 R5 P6	I ₁₁	(TYPICAL RGB 888) Green (bit weight 1) Green (bit weight 2) Green (bit weight 4) Green (bit weight 8) Green (bit weight 16) Green (bit weight 32) Green (bit weight 64) Green (bit weight 128)
PDATA_16 PDATA_17 PDATA_18 PDATA_19 PDATA_20 PDATA_21 PDATA_22 PDATA_23	R6 P7 R7 P8 R8 P9 R9 P10	I ₁₁	(TYPICAL RGB 888) Red (bit weight 1) Red (bit weight 2) Red (bit weight 4) Red (bit weight 8) Red (bit weight 16) Red (bit weight 32) Red (bit weight 64) Red (bit weight 128)
3DR	N6	I ₁₁	3D reference - For 3D applications: Left or right 3D reference (left = 1, right = 0). To be provided by the host when a 3D command is not provided. Must transition in the middle of each frame (no closer than 1 ms to the active edge of VSNC). - For light control applications: 3DR on master controller is reserved for TRIG_IN. Active high or low pulse that displays the next pattern in internal streaming mode. - The 3DR pin of the Slave Controller should be connected to the TSTPT_2 signal of the Master Controller. Based on VINTF a voltage translation might be necessary.

(1) PDATA(23:0) bus mapping is pixel format and source mode dependent. See later sections for details.

(2) PDM_CVS_TE is optional for parallel interface operation. If unused, ground the inputs or pull them down to ground through an external resistor (8 kΩ or less).

(3) Pixel clock capture edge is software programmable.

(4) The parallel data mask signal input is optional for parallel interface operations. If unused, ground the inputs or pull them down to ground through an external resistor (8 kΩ or less).

(5) VSNC, HSNC, and DATAEN polarity is software programmable.

Pin Functions – DMD Reset and Bias Control

PIN		I/O	DESCRIPTION
NAME	NUMBER		
DMD_DEN_ARSTZ	B1	O ₂	DMD driver enable (active high)/DMD reset (active low). Assuming the corresponding I/O power is supplied, this signal will be driven low after the DMD is parked and before power is removed from the DMD. If the 1.8-V power to the DLPC3479 is independent of the 1.8-V power to the DMD, then TI recommends a weak, external pulldown resistor to hold the signal low in the event the DLPC3479 power is inactive while the DMD power is applied.
DMD_LS_CLK	A1	O ₃	DMD, low speed interface clock
DMD_LS_WDATA	A2	O ₃	DMD, low speed serial write data
DMD_LS_RDATA	B2	I ₆	DMD, low speed serial read data

Pin Functions – DMD Sub-LVDS Interface

PIN		I/O	DESCRIPTION
NAME	NUMBER		
DMD_HS_CLK_P DMD_HS_CLK_N	A7 B7	O ₄	DMD high speed interface
DMD_HS_WDATA_H_P DMD_HS_WDATA_H_N DMD_HS_WDATA_G_P DMD_HS_WDATA_G_N DMD_HS_WDATA_F_P DMD_HS_WDATA_F_N DMD_HS_WDATA_E_P DMD_HS_WDATA_E_N DMD_HS_WDATA_D_P DMD_HS_WDATA_D_N DMD_HS_WDATA_C_P DMD_HS_WDATA_C_N DMD_HS_WDATA_B_P DMD_HS_WDATA_B_N DMD_HS_WDATA_A_P DMD_HS_WDATA_A_N	A3 B3 A4 B4 A5 B5 A6 B6 A8 B8 A9 B9 A10 B10 A11 B11	O ₄	DMD high speed interface lanes, write data bits: The true numbering and application of the DMD_HS_DATA pins are software configuration dependent

Pin Functions – Peripheral Interface⁽¹⁾

PIN		I/O	DESCRIPTION
NAME	NUMBER		
CMP_OUT	A12	I ₆	Successive approximation ADC comparator output (DLPC3479 Input). Assumes a successive approximation ADC is implemented with a WPC light sensor and/or a thermistor feeding one input of an external comparator and the other side of the comparator is driven from the ASIC's CMP_PWM pin. Pull down to ground if this function is not used. Hysteresis buffer.
CMP_PWM	A15	O ₁	Successive approximation comparator pulse-duration modulation (output). Supplies a PWM signal to drive the successive approximation ADC comparator used in WPC light-to-voltage sensor applications. Leave this pin unconnected, if this function is not used.
HOST_IRQ ⁽²⁾	N8	O ₉	Host interrupt (output) HOST_IRQ indicates when DLPC3479 auto-initialization is in progress and most importantly when it completes. The DLPC3479 tri-states this output during reset and assumes that an external pullup is in place to drive this signal to its inactive state.
IIC0_SCL	N10	B ₇	I ² C slave (Port 0) SCL (Bidirectional, open-drain signal with input hysteresis): An external pullup is required. The slave I ² C I/Os are 3.6-V tolerant (high-volt-input tolerant) and are powered by V _{CC_INTF} (which can be 1.8, 2.5, or 3.3 V). External I ² C pullups must be connected to an equal or higher supply voltage, up to a maximum of 3.6 V (a lower pullup supply voltage would not likely satisfy the V _{IH} specification of the slave I ² C input buffers).
IIC1_SCL	R11	B ₈	I ² C master (Port 1) SCL (Bidirectional, open-drain signal with input hysteresis): An external pull-up is required. The master I ² C I/Os are 3.6-V tolerant (High-volt-input tolerant) but are powered by the fixed 1.8-V V _{CC18} supply. Thus even though its supply is fixed at 1.8V, the external I ² C pull-ups can be connected to 1.8-V, 2.5-V, or 3.3-V supplies. However this is only valid if the external Slave I ² C device satisfies the DLPC3479 V _{IH} input requirement at the fixed 1.8-V level. V _{IH} is specified at 1.17 V, thus assuming V _{CC18} = 1.64 V, V _{IH} = 0.486 V.
IIC1_SDA	R10	B ₈	I ² C master (Port 1) SDA (Bidirectional, open-drain signal with input hysteresis): An external pull-up is required. The master I ² C I/Os are 3.6 V tolerant (High-volt-input tolerant) but are powered by the fixed 1.8-V V _{CC18} supply. Thus even though its supply is fixed at 1.8 V, the external I ² C pull-ups can be connected to 1.8-V, 2.5-V, or 3.3-V supplies; However this is only valid if the external Slave I ² C device satisfies the DLPC3479 V _{IH} input requirement at the fixed 1.8-V level. V _{IH} is specified at 1.17 V, thus assuming V _{CC18} = 1.64 V, V _{IH} = 0.486 V.
Reserved	R11	B ₈	TI internal use. TI recommends an external pullup resistor.
IIC0_SDA	N9	B ₇	I ² C slave (Port 0) SDA (Bidirectional, open-drain signal with input hysteresis): An external pullup is required. The slave I ² C port is the control port of ASIC. The slave I ² C I/Os are 3.6-V tolerant (high-volt-input tolerant) and are powered by V _{CC_INTF} (which can be 1.8, 2.5, or 3.3 V). External I ² C pullups must be connected to an equal or higher supply voltage, up to a maximum of 3.6 V (a lower pullup supply voltage would not likely satisfy the V _{IH} specification of the slave I ² C input buffers).
Reserved	R10	B ₈	TI internal use. TI recommends an external pullup resistor.
LED_SEL_0	B15	O ₁	LED enable select. Controlled by programmable DMD sequence Timing LED_SEL(1:0) 00 Enabled LED 01 DLPA3000 or DLPA3005 device application 10 None 11 Red Green Blue
LED_SEL_1	B14	O ₁	These signals will be driven low when RESETZ is asserted and the corresponding I/O power is supplied. They will continue to be driven low throughout the auto-initialization process. A weak, external pulldown resistor is still recommended to ensure that the LEDs are disabled when I/O power is not applied.
SPI0_CLK	A13	O ₁₃	Synchronous serial port 0, clock
SPI0_CSZ0	A14	O ₁₃	SPI port 1, chip select 0 (active low output) TI recommends an external pullup resistor to avoid floating inputs to the external SPI device during ASIC reset assertion.
SPI0_CSZ1	C12	O ₁₃	SPI port 1, chip select 1 (active low output) TI recommends an external pullup resistor to avoid floating inputs to the external SPI device during ASIC reset assertion.
SPI0_DIN	B12	I ₁₂	Synchronous serial port 0, receive data in
SPI0_DOUT	B13	O ₁₃	Synchronous serial port 0, transmit data out

(1) External pullup resistor must be 8 kΩ or less.

(2) For more information about usage, see [HOST_IRQ Usage Model](#).

Pin Functions – GPIO Peripheral Interface⁽¹⁾

PIN		I/O	DESCRIPTION ⁽²⁾
NAME	NUMBER		
GPIO_19	M15	B ₁	HBT_ODAT (Output): Required connection for dual ASIC applications. Connect this to alternate ASICs HBT_IDAT Input pin.
GPIO_18	M14	B ₁	HBT_OCLK (Output): Required connection for dual ASIC applications. Connect this to alternate ASICs HBT_ICLK Input pin.
GPIO_17	L15	B ₁	HBT_IDAT (Input): Required connection for dual ASIC applications. Connect this to alternate ASICs HBT_ODAT Output pin.
GPIO_16	L14	B ₁	HBT_ICLK (Input): Required connection for dual ASIC applications. Connect this to alternate ASICs HBT_OCLK Output pin.
GPIO_15	K15	B ₁	DA_SYNC (BiDir): Required to be connected between ASICs on this same pin for dual ASIC applications.
GPIO_14	K14	B ₁	SEQ_SYNC (BiDir): Required to be connected between ASICs on this same pin, with an external pull-up resistor, for dual ASIC applications.
GPIO_13	J15	B ₁	General purpose I/O 13 (hysteresis buffer). Options: 1. CAL_PWR (output): Intended to feed the calibration control of the successive approximation ADC light sensor. 2. Optional GPIO. Configure as a logic zero GPIO output and leave unconnected if not used (otherwise it will require an external pullup or pulldown to avoid a floating GPIO input).
GPIO_12	J14	B ₁	General purpose I/O 12 (hysteresis buffer). Options: 1. (Output) power enable control for LABB light sensor. 2. Optional GPIO. Configure as a logic zero GPIO output and leave unconnected if not used (otherwise it will require an external pullup or pulldown to avoid a floating GPIO input).
GPIO_11	H15	B ₁	General purpose I/O 11 (hysteresis buffer). Options: 1. Output: Thermistor power enable. 2. Optional GPIO. Configure as a logic zero GPIO output and leave unconnected if not used (otherwise it will require an external pullup or pulldown to avoid a floating GPIO input).
GPIO_10	H14	B ₁	General Purpose I/O 10 (hysteresis buffer). Options: 1. RC_CHARGE (output): Intended to feed the RC charge circuit of the successive approximation ADC used to control the light sensor comparator. 2. Optional GPIO. Configure as a logic zero GPIO output and leave unconnected if not used (otherwise it will require an external pullup or pulldown to avoid a floating GPIO input).
GPIO_09	G15	B ₁	General purpose I/O 09 (hysteresis buffer). Options: 1. LS_PWR (active high output): Intended to feed the power control signal of the successive approximation ADC light sensor. 2. 3D Glasses Control (Output): Intended to be used to control the shutters on 3D Glasses (Left = 1, Right = 0). 3. Optional GPIO. Configure as a logic zero GPIO output and leave unconnected if not used (otherwise it will require an external pullup or pulldown to avoid a floating GPIO input).
GPIO_08	G14	B ₁	General purpose I/O 08 (hysteresis buffer). Options: 1. (All) Normal mirror parking request (active low): To be driven by the PROJ_ON output of the host. A logic low on this signal will cause the DLPC3479 to PARK the DMD, but it will not power down the DMD (the DLPA3000 or DLPA3005 device does that instead). The minimum high time is 200 ms. The minimum low time is also 200 ms.
GPIO_07	F15	B ₁	General purpose I/O 07 (hysteresis buffer). Options: 1. (All) LED_ENABLE (active high input). This signal can be used as an optional shutdown interlock for the LED driver. Specifically, when so configured, setting LED_ENABLE = 0 (disabled), will cause LDEDRV_ON to be forced to 0 and LED_SEL(2:0) to be forced to b000. Otherwise when LED_ENABLE = 1 (enabled), the ASIC is free to control the LED SEL signals as it desires. There is however a 100-ms delay after LED_ENABLE transitions from low-to-high before the interlock is released. 2. Display: LABB output sample and hold sensor control signal (Output). 3. Light Control: GPIO_07 on master controller is reserved for TRIG_OUT_2 signal (Output). 4. Optional GPIO. Configure as a logic zero GPIO output and leave unconnected if not used (otherwise it will require an external pullup or pulldown to avoid a floating GPIO input).

(1) GPIO signals must be configured through software for input, output, bidirectional, or open-drain. Some GPIO have one or more alternative use modes, which are also software configurable. The reset default for all GPIO is as an input signal. An external pullup is required for each signal configured as open-drain.

(2) DLPC3479 general purpose I/O. These GPIO are software configurable.

Pin Functions – GPIO Peripheral Interface⁽¹⁾ (continued)

PIN		I/O	DESCRIPTION ⁽²⁾
NAME	NUMBER		
GPIO_06	F14	B ₁	General purpose I/O 06 (hysteresis buffer). Option: 1. Light Control: GPIO_06 on master controller is reserved for Pattern Ready Signal, which is configurable output active high or low pulse that indicates the first pattern in a pattern sequence. 2. Optional GPIO. Configure as a logic zero GPIO output and leave unconnected if not used (otherwise it will require an external pullup or pulldown to avoid a floating GPIO input).
GPIO_05	E15	B ₁	General purpose I/O 05 (hysteresis buffer). Options: 1. • GPIO_5 from the Master Controller should be connected to GPIO_6 on the Slave Controller.
GPIO_04	E14	B ₁	MST_SLVZ (Input): Master/Slave ASIC identifier strap (Master = 1, Slave = 0).
GPIO_03	D15	B ₁	General purpose I/O 03 (hysteresis buffer). Options: 1. SPI1_CSZ0 (active low output): Optional SPI1 chip select 0 signal. An external pullup resistor is required to deactivate this signal during reset and auto-initialization processes. 2. Optional GPIO. Configure as a logic zero GPIO output and leave unconnected if not used (otherwise it will require an external pullup or pulldown to avoid a floating GPIO input).
GPIO_02	D14	B ₁	General purpose I/O 02 (hysteresis buffer). Options: 1. SPI1_DOUT (output): Optional SPI1 data output signal. 2. Optional GPIO. Configure as a logic zero GPIO output and leave unconnected if not used (otherwise it will require an external pullup or pulldown to avoid a floating GPIO input).
GPIO_01	C15	B ₁	General purpose I/O 01 (hysteresis buffer). Options: 1. SPI1_CLK (output): Optional SPI1 clock signal. 2. Optional GPIO. Configure as a logic zero GPIO output and leave unconnected if not used (otherwise it will require an external pullup or pulldown to avoid a floating GPIO input).
GPIO_00	C14	B ₁	General purpose I/O 00 (hysteresis buffer). Options: 1. SPI1_DIN (input): Optional SPI1 data input signal. 2. Optional GPIO. Configure as a logic zero GPIO output and leave unconnected if not used (otherwise it will require an external pullup or pulldown to avoid a floating GPIO input).

Pin Functions – Clock and PLL Support

PIN		I/O	DESCRIPTION
NAME	NUMBER		
PLL_REFCLK_I	H1	I ₁₁	Reference clock crystal input. Use this pin as the oscillator input if an external oscillator is used in place of a crystal.
PLL_REFCLK_O	J1	O ₅	Reference clock crystal return. If an external oscillator is used in place of a crystal, leave this pin unconnected (that is floating with no added capacitive load).

Pin Functions – Power and Ground⁽¹⁾

PIN		I/O	DESCRIPTION
NAME	NUMBER		
V _{DD}	C5, D5, D7, D12, J4, J12, K3, L4, L12, M6, M9, D9, D13, F13, H13, L13, M10, D3, E3	PWR	Core power 1.1 V (main 1.1 V)
V _{DDL12}	C3	PWR	Core power 1.1 V
V _{SS}	Common to all package types C4, D6, D8, D10, E4, E13, F4, G4, G12, H4, H12, J3, J13, K4, K12, L3, M4, M5, M8, M12, G13, C6, C8 Only available on DLPC3479 F6, F7, F8, F9, F10, G6, G7, G8, G9, G10, H6, H7, H8, H9, H10, J6, J7, J8, J9, J10, K6, K7, K8, K9, K10	GND	Core ground (eDRAM, I/O ground, thermal ground)
V _{CC18}	C7, C9, D4, E12, F12, K13, M11	PWR	All 1.8-V I/O power: (1.8-V power supply for all I/O other than the host or parallel interface and the SPI flash interface. This includes RESETZ, PARKZ LED_SEL, CMP, GPIO, IIC1, TSTPT, and JTAG pins)
V _{CC_INTF}	M3, M7, N3, N7	PWR	Host or parallel interface I/O power: 1.8 to 3.3 V (Includes IIC0, PDATA, video syncs, and HOST_IRQ pins)
V _{CC_FLASH}	D11	PWR	Flash interface I/O power: 1.8 to 3.3 V (Dedicated SPI0 power pin)
V _{DD_PLLM}	H2	PWR	MCG PLL 1.1-V power
V _{SS_PLLM}	G3	RTN	MCG PLL return
V _{DD_PLLD}	J2	PWR	DCG PLL 1.1-V power
V _{SS_PLLD}	H3	RTN	DCG PLL return

(1) The only power sequencing restrictions are:

- The VDDL12 supply must be powered-on at exactly the same time or after the VDD11 supply.
- The VDD11 supply ramps up with a 1-ms minimum rise time.
- The reverse is needed at power down.

Table 1. I/O Type Subscript Definition

SUBSCRIPT	I/O	SUPPLY REFERENCE	ESD STRUCTURE
	DESCRIPTION		
1	1.8-V LVCMOS I/O buffer with 8-mA drive	V _{CC18}	ESD diode to GND and supply rail
2	1.8-V LVCMOS I/O buffer with 4-mA drive	V _{CC18}	ESD diode to GND and supply rail
3	1.8-V LVCMOS I/O buffer with 24-mA drive	V _{CC18}	ESD diode to GND and supply rail
4	1.8-V sub-LVDS output with 4-mA drive	V _{CC18}	ESD diode to GND and supply rail
5	1.8-, 2.5-, 3.3-V LVCMOS with 4-mA drive	V _{CC_INTF}	ESD diode to GND and supply rail
6	1.8-V LVCMOS input	V _{CC18}	ESD diode to GND and supply rail
7	1.8-, 2.5-, 3.3-V I ² C with 3-mA drive	V _{CC_INTF}	ESD diode to GND and supply rail
8	1.8-V I ² C with 3-mA drive	V _{CC18}	ESD diode to GND and supply rail
9	1.8-, 2.5-, 3.3-V LVCMOS with 8-mA drive	V _{CC_INTF}	ESD diode to GND and supply rail
11	1.8, 2.5, 3.3-V LVCMOS input	V _{CC_INTF}	ESD diode to GND and supply rail
12	1.8-, 2.5-, 3.3-V LVCMOS input	V _{CC_FLSH}	ESD diode to GND and supply rail
13	1.8-, 2.5-, 3.3-V LVCMOS with 8-mA drive	V _{CC_FLSH}	ESD diode to GND and supply rail

Table 2. Internal Pullup and Pulldown Characteristics⁽¹⁾⁽²⁾

INTERNAL PULLUP AND PULLDOWN RESISTOR CHARACTERISTICS	VCCIO	MIN	MAX	UNIT
Weak pullup resistance	3.3 V	29	63	kΩ
	2.5 V	38	90	kΩ
	1.8 V	56	148	kΩ
Weak pulldown resistance	3.3 V	30	72	kΩ
	2.5 V	36	101	kΩ
	1.8 V	52	167	kΩ

(1) The resistance is dependent on the supply voltage level applied to the I/O.

(2) An external 8-kΩ pullup or pulldown (if needed) would work for any voltage condition to correctly pull enough to override any associated internal pullups or pulldowns.

6 Specifications

6.1 Absolute Maximum Ratings⁽¹⁾

over operating free-air temperature (unless otherwise noted)

		MIN	MAX	UNIT
SUPPLY VOLTAGE⁽²⁾⁽³⁾				
$V_{(VDD)}$ (core)		–0.3	1.21	V
$V_{(VDDL12)}$ (core)		–0.3	1.32	V
Power + sub-LVDS		–0.3	1.96	V
$V_{(VCC_INTF)}$	Host I/O power	–0.3	3.60	V
	If 1.8-V power used	–0.3	1.99	
	If 2.5-V power used	–0.3	2.75	
	If 3.3-V power used	–0.3	3.60	
$V_{(VCC_FLSH)}$	Flash I/O power	–0.3	3.60	V
	If 1.8-V power used	–0.3	1.96	
	If 2.5-V power used	–0.3	2.72	
	If 3.3-V power used	–0.3	3.58	
$V_{(VDD_PLLM)}$ (MCG PLL)		–0.3	1.21	V
$V_{(VDD_PLLD)}$ (1DCG PLL)		–0.3	1.21	V
GENERAL				
T_J	Operating junction temperature	–30	125	°C
T_{stg}	Storage temperature	–40	125	°C

- (1) Stresses beyond those listed under *Absolute Maximum Ratings* may cause permanent damage to the device. These are stress ratings only, which do not imply functional operation of the device at these or any other conditions beyond those indicated under *Recommended Operating Conditions*. Exposure to absolute-maximum-rated conditions for extended periods may affect device reliability.
- (2) All voltage values are with respect to GND.
- (3) Overlap currents, if allowed to continue flowing unchecked, not only increase total power dissipation in a circuit, but degrade the circuit reliability, thus shortening its usual operating life.

6.2 ESD Ratings

		VALUE	UNIT
$V_{(ESD)}$ ⁽¹⁾ Electrostatic discharge	Human body model (HBM), per ANSI/ESDA/JEDEC JS-001, all pins ⁽²⁾	2000	V
	Charged device model (CDM), per JEDEC specification JESD22-C101, all pins ⁽³⁾	–500	

- (1) Electrostatic discharge (ESD) to measure device sensitivity and immunity to damage caused by assembly line electrostatic discharges in to the device.
- (2) JEDEC document JEP155 states that 500-V HBM allows safe manufacturing with a standard ESD control process.
- (3) JEDEC document JEP157 states that 250-V CDM allows safe manufacturing with a standard ESD control process.

6.3 Recommended Operating Conditions

over operating free-air temperature range (unless otherwise noted)

			MIN	NOM	MAX	UNIT
V _(VDD)	Core power 1.1 V (main 1.1 V)	±5% tolerance	1.045	1.1	1.155	V
V _(VDDL12)	Core power 1.1 V	±5% tolerance See ⁽¹⁾	1.02	1.1	1.18	V
			1.12	1.2	1.28	
V _(VCC18)	All 1.8-V I/O power: (1.8-V power supply for all I/O other than the host or parallel interface and the SPI flash interface. This includes RESETZ, PARKZ LED_SEL, CMP, GPIO, IIC1, TSTPT, and JTAG pins.)	±8.5% tolerance	1.64	1.8	1.96	V
V _(VCC_INTF)	Host or parallel interface I/O power: 1.8 to 3.3 V (includes IIC0, PDATA, video syncs, and HOST_IRQ pins)	±8.5% tolerance See ⁽¹⁾	1.64	1.8	1.96	V
			2.28	2.5	2.72	
			3.02	3.3	3.58	
V _(VCC_FLASH)	Flash interface I/O power: 1.8 to 3.3 V	±8.5% tolerance See ⁽¹⁾	1.64	1.8	1.96	V
			2.28	2.5	2.72	
			3.02	3.3	3.58	
V _(VDD_PLLM)	MCG PLL 1.1-V power	±9.1% tolerance See ⁽²⁾	1.025	1.1	1.155	V
V _(VDD_PLLD)	DCG PLL 1.1-V power	±9.1% tolerance See ⁽²⁾	1.025	1.1	1.155	V
T _A	Operating ambient temperature range ⁽³⁾		–30		85	°C
T _J	Operating junction temperature		–30		105	°C

(1) These supplies have multiple valid ranges.

(2) These I/O supply ranges are wider to facilitate additional filtering.

(3) The operating ambient temperature range assumes 0 forced air flow, a JEDEC JESD51 junction-to-ambient thermal resistance value at 0 forced air flow @_{θJA} at 0 m/s), a JEDEC JESD51 standard test card and environment, along with minimum and maximum estimated power dissipation across process, voltage, and temperature. Thermal conditions vary by application, which will impact R_{θJA}. Thus, maximum operating ambient temperature varies by application.

(a) $T_{a_min} = T_{j_min} - (P_{d_min} \times R_{\theta JA}) = -30^{\circ}\text{C} - (0.0\text{ W} \times 30.3^{\circ}\text{C/W}) = -30^{\circ}\text{C}$

(b) $T_{a_max} = T_{j_max} - (P_{d_max} \times R_{\theta JA}) = 105^{\circ}\text{C} - (0.348\text{ W} \times 30.3^{\circ}\text{C/W}) = +94.4^{\circ}\text{C}$

6.4 Thermal Information

THERMAL METRIC ⁽¹⁾		DLPC3479	UNIT
		ZEZ (NFBGA)	
		201 PINS	
R _{θJC}	Junction-to-case thermal resistance	10.1	°C/W
R _{θJA} ⁽²⁾	Junction-to-air thermal resistance	At 0 m/s of forced airflow	28.8
		At 1 m/s of forced airflow	25.3
		At 2 m/s of forced airflow	24.4
ψ _{JT} ⁽³⁾	Temperature variance from junction to package top center temperature, per unit power dissipation	0.23	°C/W

(1) For more information about traditional and new thermal metrics, see the *Semiconductor and IC Package Thermal Metrics* application report, [SPRA953](#).

(2) Thermal coefficients abide by JEDEC Standard 51. R_{θJA} is the thermal resistance of the package as measured using a JEDEC defined standard test PCB. This JEDEC test PCB is not necessarily representative of the DLPC3479 test PCB and thus the reported thermal resistance may not be accurate in the actual product application. Although the actual thermal resistance may be different, it is the best information available during the design phase to estimate thermal performance.

(3) Example: (0.5 W) × (0.2°C/W) ≈ 1.00°C temperature rise.

6.5 Electrical Characteristics over Recommended Operating Conditions

see (1)(2)(3)(4)

	PARAMETER	TEST CONDITIONS ⁽⁵⁾⁽⁶⁾	MIN	TYP ⁽⁷⁾	MAX ⁽⁸⁾	UNIT
$V_{(VDD)}$	Core current 1.1 V (main 1.1 V)	IDLE disabled, 1920 × 1080, 60 Hz		188	334	mA
$V_{(VDD_PLL M)}$	MCG PLL 1.1 V-current	IDLE disabled, 1920 × 1080, 60 Hz		4	7	mA
$V_{(VDD_PLL D)}$	DCG PLL 1.1 V-current	IDLE disabled, 1920 × 1080, 60 Hz		4	7	mA
$V_{(VDD)} + V_{(VDD_PLL M)} + V_{(VDD_PLL D)}$	Core Current 1.1 V + MCG PLL 1.1-V current + DCG PLL 1.1-V current	IDLE disabled, 1920 × 1080, 60 Hz		196	348	mA
$V_{(VCC18)}$	Main 1.8-V I/O current: 1.8-V power supply for all I/O other than the host or parallel interface and the SPI flash interface. This includes sub-LVDS DMD I/O, RESETZ, PARKZ, LED_SEL, CMP, GPIO, IIC1, TSTPT and JTAG pins	IDLE disabled, 1920 × 1080, 60 Hz			62	mA
$V_{(VCC_INTF)}$	Host or parallel interface I/O current: 1.8 to 3.3 V (includes IIC0, PDATA, video syncs, and HOST_IRQ pins)	IDLE disabled, 1920 × 1080, 60 Hz			3	mA
$V_{(VCC_FLSH)}$	Flash Interface I/O current: 1.8 to 3.3 V	IDLE disabled, 1920 × 1080, 60 Hz		1	1.5	mA
$V_{(VCC18)} + V_{(VCC_INTF)} + V_{(VCC_FLSH)}$	Main 1.8 V I/O current + V_{CC_INTF} current + V_{CC_FLSH} current	IDLE disabled, 1920 × 1080, 60 Hz		30	66.5	mA

- (1) Assumes 12.5% activity factor, 30% clock gating on appropriate domains, and mixed SVT or HVT cells.
- (2) Programmable host and flash I/O are at minimum voltage (that is 1.8 V) for this typical scenario.
- (3) Max currents column use typical motion video as the input. The typical currents column uses SMPTE color bars as the input.
- (4) Some applications may be forced to use 1-oz copper to manage ASIC package heat.
- (5) Input image is 1920 × 1080 (1080p) 24-bits on the parallel interface at the frame rate shown with a 0.47-inch 1080p DMD.
- (6) In normal operation while displaying an image with CAIC enabled.
- (7) Assumes typical case power PVT condition = nominal process, typical voltage, typical temperature (55°C junction), a 0.47-inch 1080p DMD.
- (8) Assumes worse case power PVT condition = corner process, high voltage, high temperature (105°C junction), a 0.47-inch 1080p DMD.

6.6 Electrical Characteristics

over operating free-air temperature range (unless otherwise noted)⁽¹⁾⁽²⁾

PARAMETER ⁽³⁾		TEST CONDITIONS	MIN	TYP	MAX	UNIT
V_{IH}	High-level input threshold voltage	I ² C buffer (I/O type 7)	$0.7 \times V_{CC_INTF}$		⁽¹⁾	V
		1.8-V LVTTL (I/O type 1, 2, 3, 5, 6, 8, 9, 11, 12, 13)	1.17		3.6	
		1.8-V LVTTL (I/O type 1, 6) identified below: (2) CMP_OUT; PARKZ; RESETZ; GPIO_0 → 19	1.3		3.6	
		2.5-V LVTTL (I/O type 5, 9, 11, 12, 13)	1.7		3.6	
		3.3-V LVTTL (I/O type 5, 9, 11, 12, 13)	2		3.6	
V_{IL}	Low-level input threshold voltage	I ² C buffer (I/O type 7)	−0.5		$0.3 \times V_{CC_INTF}$	V
		1.8-V LVTTL (I/O type 1, 2, 3, 5, 6, 8, 9, 11, 12, 13)	−0.3		0.63	
		1.8-V LVTTL (I/O type 1, 6) identified below: (2) CMP_OUT; PARKZ; RESETZ; GPIO_00 through GPIO_19	−0.3		0.5	
		2.5-V LVTTL (I/O type 5, 9, 11, 12, 13)	−0.3		0.7	
		3.3-V LVTTL (I/O type 5, 9, 11, 12, 13)	−0.3		0.8	
V_{CM}	Steady-state common mode voltage	1.8-V sub-LVDS (DMD high speed) (I/O type 4)	0.8	0.9	1	mV
I_{VODI}	Differential output magnitude	1.8-V sub-LVDS (DMD high speed) (I/O type 4)	200			mV
V_{OH}	High-level output voltage	1.8-V LVTTL (I/O type 1, 2, 3, 5, 6, 8, 9, 11, 12, 13)	1.35			V
		2.5-V LVTTL (I/O type 5, 9, 11, 12, 13)	1.7			
		3.3-V LVTTL (I/O type 5, 9, 11, 12, 13)	2.4			
		1.8-V sub-LVDS – DMD high speed (I/O type 4)	1			
V_{OL}	Low-level output voltage	I ² C buffer (I/O type 7)	$V_{CC_INTF} > 2\text{ V}$		0.4	V
		I ² C buffer (I/O type 7)	$V_{CC_INTF} < 2\text{ V}$		$0.2 \times V_{CC_INTF}$	
		1.8-V LVTTL (I/O type 1, 2, 3, 5, 6, 8, 9, 11, 12, 13)			0.45	
		2.5-V LVTTL (I/O type 5, 9, 11, 12, 13)			0.7	
		3.3-V LVTTL (I/O type 5, 9, 11, 12, 13)			0.4	
		1.8-V sub-LVDS – DMD high speed (I/O type 4)	0.8			

(1) I/O is high voltage tolerant; that is, if $V_{CC} = 1.8\text{ V}$, the input is 3.3-V tolerant, and if $V_{CC} = 3.3\text{ V}$, the input is 5-V tolerant.

(2) ASIC pins: CMP_OUT; PARKZ; RESETZ; GPIO_00 through GPIO_19 have slightly varied V_{IH} and V_{IL} range from other 1.8-V I/O.

(3) The number inside each parenthesis for the I/O refers to the type defined in [Table 1](#).

Electrical Characteristics (continued)

 over operating free-air temperature range (unless otherwise noted)⁽¹⁾⁽²⁾

PARAMETER ⁽³⁾		TEST CONDITIONS	MIN	TYP	MAX	UNIT
I_{OH}	High-level output current	1.8-V LVTTL (I/O type 2, 4, 5)		2		mA
		1.8-V LVTTL (I/O type 1, 9, 13)		3.5		
		1.8-V LVTTL (I/O type 3)		10.6		
		2.5-V LVTTL (I/O type 5)		5.4		
		2.5-V LVTTL (I/O type 9, 13)		10.8		
		2.5-V LVTTL (I/O type 5, 9, 11, 12, 13)		28.7		
		3.3-V LVTTL (I/O type 5)		7.8		
		3.3-V LVTTL (I/O type 9, 13)		15		
I_{OL}	Low-level output current	I ² C buffer (I/O type 7)		3		mA
		1.8-V LVTTL (I/O type 2, 4, 5)		2.3		
		1.8-V LVTTL (I/O type 1, 9, 13)		4.6		
		1.8-V LVTTL (I/O type 3)		13.9		
		2.5-V LVTTL (I/O type 5)		5.2		
		2.5-V LVTTL (I/O type 9, 13)		10.4		
		2.5-V LVTTL (I/O type 5, 9, 11, 12, 13)		31.1		
		3.3-V LVTTL (I/O type 5)		4.4		
I_{OZ}	High-impedance leakage current	I ² C buffer (I/O type 7)	$V_{I2C\text{ buffer}} < 0.1 \times V_{CC_INTF}$ or $V_{I2C\text{ buffer}} > 0.9 \times V_{CC_INTF}$	–10	10	μ A
		1.8-V LVTTL (I/O type 1, 2, 3, 5, 6, 8, 9, 11, 12, 13)		–10	10	
		2.5-V LVTTL (I/O type 5, 9, 11, 12, 13)		–10	10	
		3.3-V LVTTL (I/O type 5, 9, 11, 12, 13)		–10	10	
C_I	Input capacitance (including package)	I ² C buffer (I/O type 7)			5	pF
		1.8-V LVTTL (I/O type 1, 2, 3, 5, 6, 8, 9, 11, 12, 13)		2.6	3.5	
		2.5-V LVTTL (I/O type 5, 9, 11, 12, 13)		2.6	3.5	
		3.3-V LVTTL (I/O type 5, 9, 11, 12, 13)		2.6	3.5	
		1.8-V sub-LVDS – DMD high speed (I/O type 4)			3	

6.7 High-Speed Sub-LVDS Electrical Characteristics

over operating free-air temperature range (unless otherwise noted)

PARAMETER		MIN	NOM	MAX	UNIT
V_{CM}	Steady-state common mode voltage	0.8	0.9	1.0	V
$V_{CM}(\Delta pp)^{(1)}$	V_{CM} change peak-to-peak (during switching)			75	mV
$V_{CM}(\Delta ss)^{(1)}$	V_{CM} change steady state	–10		10	mV
$ V_{OD} ^{(2)}$	Differential output voltage magnitude		200		mV
$V_{OD}(\Delta)$	V_{OD} change (between logic states)	–10		10	mV
V_{OH}	Single-ended output voltage high		1.00		V
V_{OL}	Single-ended output voltage low		0.80		V
$t_R^{(2)}$	Differential output rise time			250	ps
$t_F^{(2)}$	Differential output fall time			250	ps
t_{MAX}	Maximum switching rate			1200	Mbps
DCout	Output duty cycle	45%	50%	55%	
$T_{Xterm}^{(1)}$	Internal differential termination	80	100	120	Ω
T_{Xload}	100- Ω differential PCB trace (50- Ω transmission lines)	0.5		6	inches

(1) For the definition of V_{CM} changes, see [Figure 2](#).

(2) Note that V_{OD} is the differential voltage swing measured across a 100- Ω termination resistance connected directly between the transmitter differential pins. $|V_{OD}|$ is the magnitude of this voltage swing relative to 0. Rise and fall times are defined for the differential V_{OD} signal as follows in [Figure 3](#).

6.8 Low-Speed SDR Electrical Characteristics

over operating free-air temperature range (unless otherwise noted)

PARAMETER	ID	TEST CONDITIONS	MIN	MAX	UNIT
Operating voltage	V _{CC18} (all signal groups)		1.64	1.96	V
DC input high voltage	V _{IHD} (DC) Signal group 1	All	$0.7 \times V_{CC18}$	$V_{CC18} + 0.5$	V
DC input low voltage ⁽¹⁾	V _{ILD} (DC) Signal group 1	All	–0.50	$0.3 \times V_{CC18}$	V
AC input high voltage ⁽²⁾	V _{IHD} (AC) Signal group 1	All	$0.8 \times V_{CC18}$	$V_{CC18} + 0.5$	V
AC input low voltage	V _{ILD} (AC) Signal group 1	All	–0.5	$0.2 \times V_{CC18}$	V
Slew rate ⁽³⁾⁽⁴⁾⁽⁵⁾⁽⁶⁾	Signal group 1		1	3.0	V/ns
	Signal group 2		0.25		
	Signal group 3		0.5		

(1) V_{ILD}(AC) minimum applies to undershoot.

(2) V_{IHD}(AC) maximum applies to overshoot.

(3) Signal group 1 output slew rate for rising edge is measured between V_{ILD}(DC) to V_{IHD}(AC).

(4) Signal group 1 output slew rate for falling edge is measured between V_{IHD}(DC) to V_{ILD}(AC).

(5) Signal group 1: See [Figure 4](#).

(6) Signal groups 2 and 3 output slew rate for rising edge is measured between V_{ILD}(AC) to V_{IHD}(AC).

6.9 System Oscillators Timing Requirements

Parameters			MIN	MAX	UNIT
f_{clk}	Clock frequency, MOSC ⁽¹⁾		23.998	24.002	MHz
t_c	Cycle time, MOSC ⁽¹⁾		41.663	41.670	ns
$t_{w(H)}$	Pulse duration, MOSC, high	50% to 50% reference points (signal)		40 $t_c\%$	
$t_{w(L)}$	Pulse duration, MOSC, low	50% to 50% reference points (signal)		40 $t_c\%$	
t_t	Transition time, MOSC, $t_t = t_f / t_r$	20% to 80% reference points (signal)		10	ns
t_{jp}	Long-term, peak-to-peak, period jitter ⁽²⁾ , MOSC (that is the deviation in period from ideal period due solely to high frequency jitter)			2	

(1) The frequency accuracy for MOSC is ± 200 PPM (This includes impact to accuracy due to aging, temperature, and trim sensitivity).

(2) The MOSC input cannot support spread spectrum clock spreading.

6.10 Power-Up and Reset Timing Requirements

			MIN	MAX	UNIT
$t_{w(L)}$	Pulse duration, inactive low, RESETZ	50% to 50% reference points (signal)	1.25		μs
t_t	Transition time, RESETZ ⁽¹⁾ , $t_t = t_f / t_r$	20% to 80% reference points (signal)		0.5	μs

(1) For more information on RESETZ, see [Pin Configuration and Functions](#).

6.11 Parallel Interface Frame Timing Requirements

			MIN	MAX	UNIT
t_{p_vsw}	Pulse duration – VSYNC_WE high	50% reference points	1		lines
t_{p_vbp}	Vertical back porch (VBP) – time from the leading edge of VSYNC_WE to the leading edge HSYNC_CS for the first active line. See ⁽¹⁾ .	50% reference points	2		lines
t_{p_vfp}	Vertical front porch (VFP) – time from the leading edge of the HSYNC_CS following the last active line in a frame to the leading edge of VSYNC_WE. See ⁽¹⁾ .	50% reference points	1		lines
t_{p_tvb}	Total vertical blanking – time from the leading edge of HSYNC_CS following the last active line of one frame to the leading edge of HSYNC_CS for the first active line in the next frame. This is equal to the sum of VBP (t_{p_vbp}) + VFP (t_{p_vfp}).	50% reference points	See ⁽¹⁾		lines
t_{p_hsw}	Pulse duration – HSYNC_CS high	50% reference points	4	128	PCLKs
t_{p_hbp}	Horizontal back porch – time from rising edge of HSYNC_CS to rising edge of DATAEN_CMD	50% reference points	4		PCLKs
t_{p_hfp}	Horizontal front porch – time from falling edge of DATAEN_CMD to rising edge of HSYNC_CS	50% reference points	8		PCLKs
t_{p_thb}	Total horizontal blanking – sum of horizontal front and back porches	50% reference points	See ⁽²⁾		PCLKs

- (1) The minimum total vertical blanking is defined by the following equation: $t_{p_tvb}(\text{min}) = 6 + [8 \times \text{Max}(1, \text{Source_ALPF} / \text{DMD_ALPF})]$ lines where:
- (a) SOURCE_ALPF = Input source active lines per frame
 - (b) DMD_ALPF = Actual DMD used lines per frame supported
- (2) Total horizontal blanking is driven by the maximum line rate for a given source which will be a function of resolution and orientation. The following equation can be applied for this: $t_{p_thb} = \text{Roundup}[(1000 \times f_{\text{clock}}) / \text{LR}] - \text{APPL}$ where:
- (a) f_{clock} = Pixel clock rate in MHz
 - (b) LR = Line rate in kHz
 - (c) APPL is the number of active pixels per (horizontal) line.
 - (d) If t_{p_thb} is calculated to be less than $t_{p_hbp} + t_{p_hfp}$ then the pixel clock rate is too low or the line rate is too high, and one or both must be adjusted.

6.12 Parallel Interface General Timing Requirements⁽¹⁾

			MIN	MAX	UNIT
f_{clock}	Clock frequency, PCLK		1.0	150.0	MHz
$t_{\text{p_clkper}}$	Clock period, PCLK	50% reference points	6.66	1000	ns
$t_{\text{p_clkjit}}$	Clock jitter, PCLK	Max f_{clock}	See ⁽²⁾	See ⁽²⁾	
$t_{\text{p_wh}}$	Pulse duration low, PCLK	50% reference points	2.43		ns
$t_{\text{p_wl}}$	Pulse duration high, PCLK	50% reference points	2.43		ns
$t_{\text{p_su}}$	Setup time – HSYNC_CS, DATEN_CMD, PDATA(23:0) valid before the active edge of PCLK	50% reference points	0.9		ns
$t_{\text{p_h}}$	Hold time – HSYNC_CS, DATEN_CMD, PDATA(23:0) valid after the active edge of PCLK	50% reference points	0.9		ns
t_{t}	Transition time – all signals	20% to 80% reference points	0.2	2.0	ns

- (1) The active (capture) edge of PCLK for HSYNC_CS, DATEN_CMD and PDATA(23:0) is software programmable, but defaults to the rising edge.
- (2) Use this formula to calculate clock jitter (in ns): Jitter = $[1 / f_{\text{clock}} - 5.76 \text{ ns}]$. Setup and hold times must be met during clock jitter.

6.13 Flash Interface Timing Requirements

The DLPC3479 ASIC flash memory interface consists of a SPI flash serial interface with a programmable clock rate. The DLPC3479 can support 1- to 64-Mb flash memories.⁽¹⁾⁽²⁾

			MIN	MAX	UNIT
f_{clock}	Clock frequency, SPI_CLK	See ⁽³⁾	1.42	36.0	MHz
$t_{\text{p_clkper}}$	Clock period, SPI_CLK	50% reference points	27.7	704	ns
$t_{\text{p_wh}}$	Pulse duration low, SPI_CLK	50% reference points	352		ns
$t_{\text{p_wl}}$	Pulse duration high, SPI_CLK	50% reference points	352		ns
t_t	Transition time – all signals	20% to 80% reference points	0.2	3.0	ns
$t_{\text{p_su}}$	Setup time – SPI_DIN valid before SPI_CLK falling edge	50% reference points	10.0		ns
$t_{\text{p_h}}$	Hold time – SPI_DIN valid after SPI_CLK falling edge	50% reference points	0.0		ns
$t_{\text{p_clqv}}$	SPI_CLK clock falling edge to output valid time – SPI_DOUT and SPI_CSZ	50% reference points		1.0	ns
$t_{\text{p_clqx}}$	SPI_CLK clock falling edge output hold time – SPI_DOUT and SPI_CSZ	50% reference points	–3.0	3.0	ns

- (1) Standard SPI protocol is to transmit data on the falling edge of SPI_CLK and capture data on the rising edge. The DLPC3479 does transmit data on the falling edge, but it also captures data on the falling edge rather than the rising edge. This provides support for SPI devices with long clock-to-Q timing. DLPC3479 hold capture timing has been set to facilitate reliable operation with standard external SPI protocol devices.
- (2) With the above output timing, DLPC3479 provides the external SPI device 8.2-ns input set-up and 8.2-ns input hold, relative to the rising edge of SPI_CLK.
- (3) This range include the 200 ppm of the external oscillator (but no jitter).

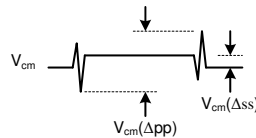
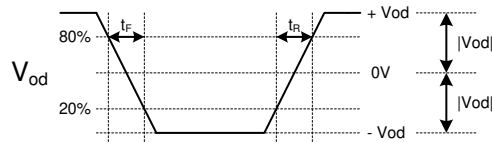


Figure 2. Definition of V_{CM} Changes



NOTE: V_{CM} is removed when the signals are viewed differentially.

Figure 3. Differential Output Signal

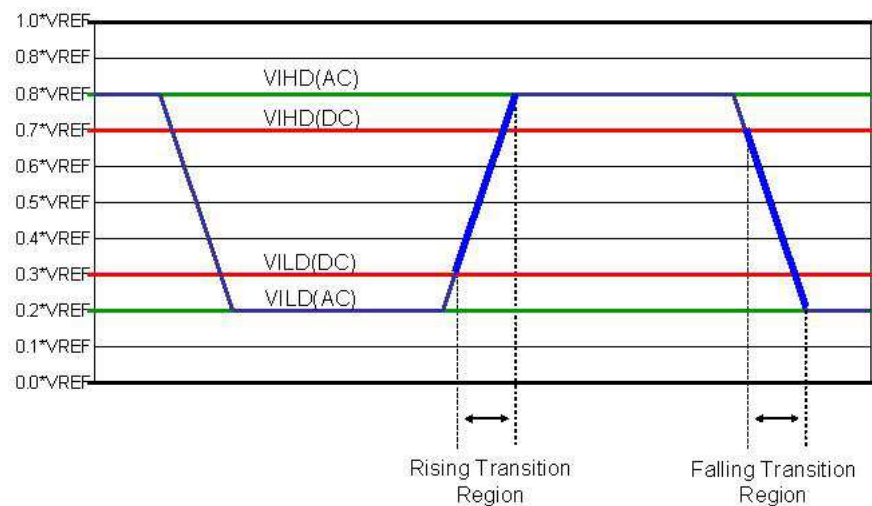


Figure 4. Low Speed (LS) I/O Input Thresholds

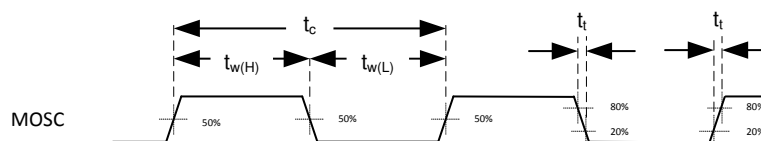


Figure 5. System Oscillators

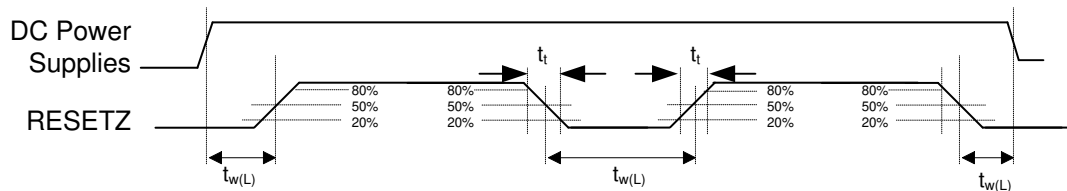
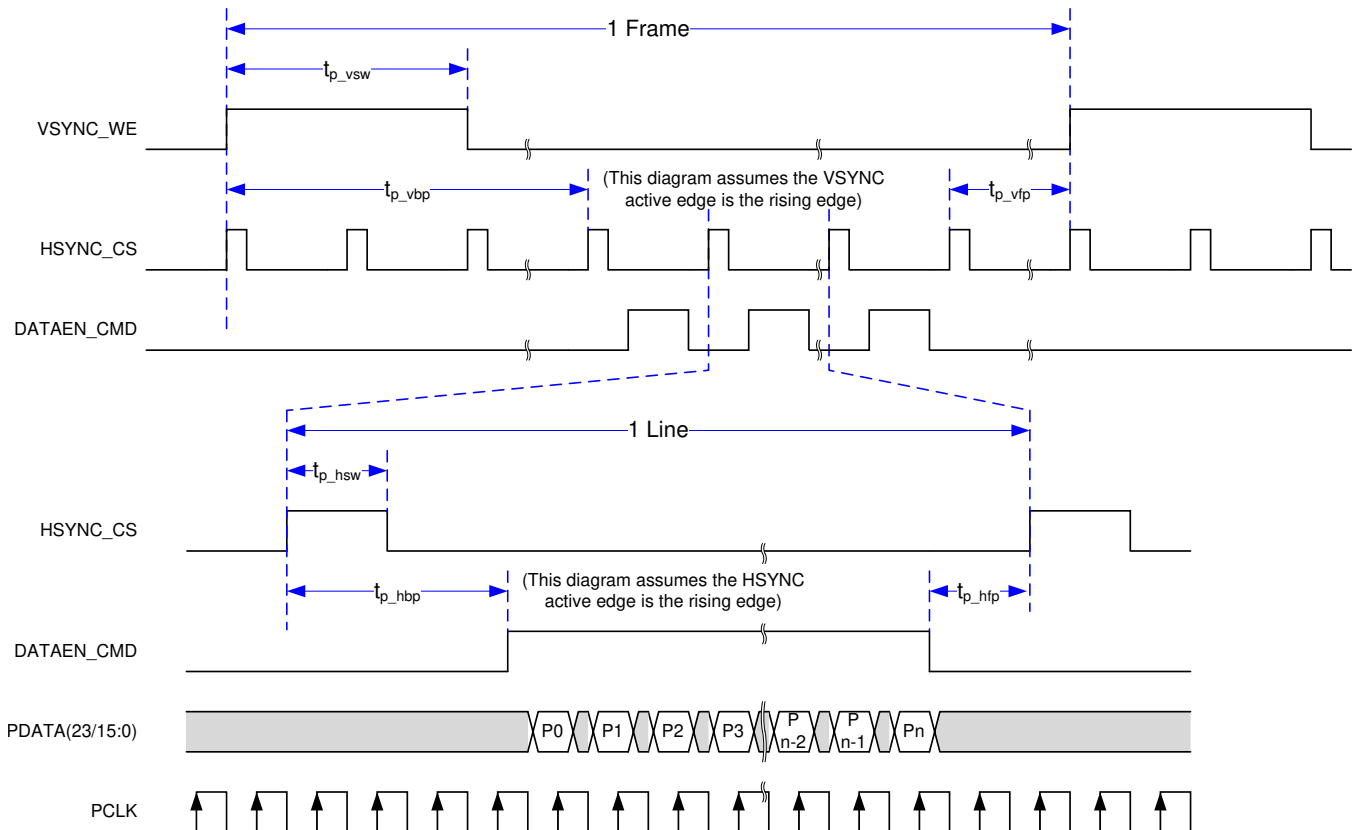
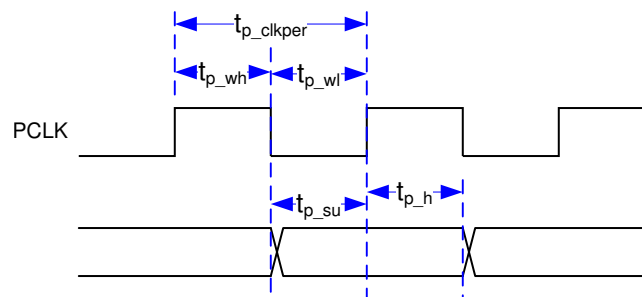


Figure 6. Power-Up and Power-Down RESETZ Timing


Figure 7. Parallel Interface Frame Timing

Figure 8. Parallel Interface General Timing

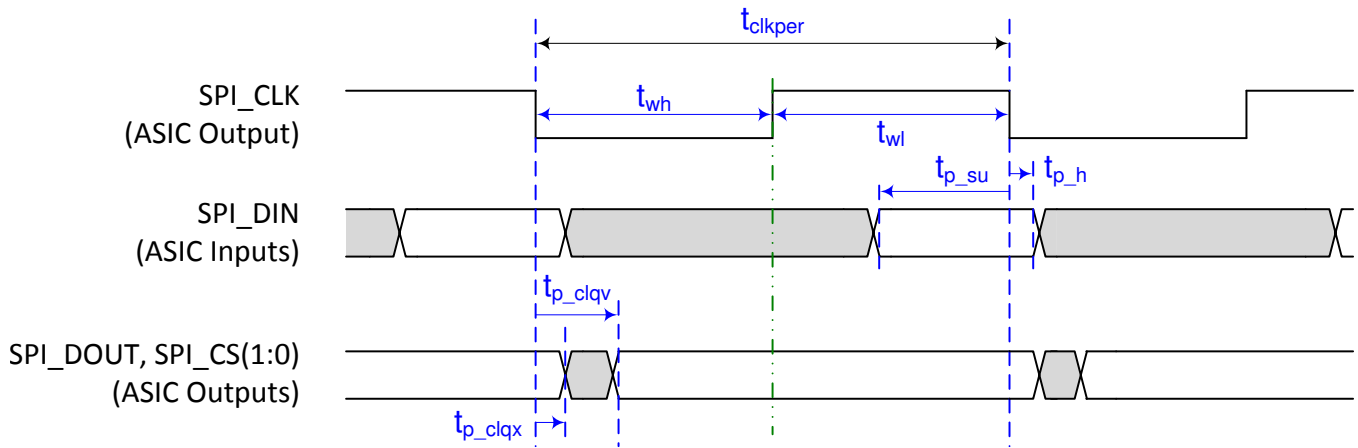
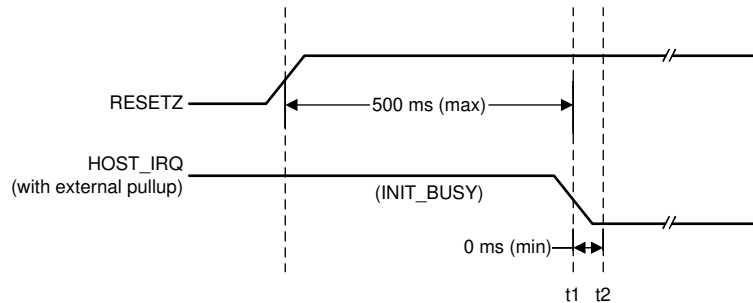


Figure 9. Flash Interface Timing

7 Parameter Measurement Information

7.1 HOST_IRQ Usage Model

- While reset is applied HOST_IRQ will reset to tri-state (an external pullup pulls the line high).
- HOST_IRQ will remain tri-state (pulled high externally) until the microprocessor boot completes. While the signal is pulled high, this indicates that the DLPC3479 is performing boot-up and auto-initialization.
- As soon as possible after boot-up, the microprocessor will drive HOST_IRQ to a logic high state to indicate that the ASIC is continuing to perform auto-initialization (no real state change occurs on the external signal).
- Upon completion of auto-initialization, software will set HOST_IRQ to a logic low state to indicate the completion of auto-initialization (At the falling edge, the system is said to enter the INIT_DONE state).
- The controller initialization will generally be completed (HOST_IRQ will go low) within 500ms of RESETZ being asserted; however, this may vary depending on the utilized software and the contents of the user configurable autoinitialization file.



t1 is the first falling edge of HOST_IRQ, At this point the auto-initiation sequence is complete.

t2 is where HOST_IRQ goes low. Ensure that I²C interface to the device does not begin until this point (within 500 ms of the release of RESETZ)

Figure 10. Host IRQ Timing

7.2 Input Source

7.2.1 Input Source - Frame Rates

The dual DLPC3479 devices support both 2D and 3D sources on the parallel interface. The DLPC3479 devices offer very limited scaling capabilities, but a limited number of input image sizes can be scaled up to 1920×1080 for displaying images on the DMD over the frame rates given below for 2D sources and 3D sources.

For 2D sources, the following two input options are supported on the parallel interface:

- 1920×1080 over 10-62 Hz
- 1280×720 over 10-62 Hz (Image scaled up to 1920×1080)

The devices have a 150 MHz maximum input clock frequency for the parallel interface which results in pixel inputs cannot be supported at 120 Hz for 1920×1080 . However, up to 120-Hz 3D can be supported if the vertical and/or horizontal resolution of the input source is reduced. The DLPC3479 devices can then scale the images up to 1920×1080 for displaying them on the DMD. For 3D sources, the following three input options are supported on the parallel interface and all are scaled up to 1920×1080 :

- 1920×540 over 98-102 Hz and 118-122 Hz
- 960×1080 over 98-102 Hz and 118-122 Hz
- 1280×720 over 98-102 Hz and 118-122 Hz

For 3D sources on the parallel interface, images must be frame-sequential (L, R, L, ...) when input to the DLPC3479 devices. Any processing required to unpack 3D images and to convert them to frame sequential must be done by external electronics prior to inputting the images to the DLPC3479 devices. Each 3D source frame input on the parallel interface must contain a single eye frame of data separated by a VSYNC where an eye frame contains image data for a single left or right eye. The signal 3DR input to the DLPC3479 devices tells whether the input frame is for the left eye or right eye.

For 3D sources, each DMD frame will be displayed at the same rate as the parallel interface frame rate. Typical timing for 50-Hz or 60-Hz 3D HDMI source frames, the parallel interface of the DLPC3479 devices, and the DMD is shown in Figure 11. GPIO_09 is optionally sent to a transmitter on the system PCB for wirelessly transmitting a synchronization signal to 3D glasses. The glasses are then in phase with the DMD images being displayed. Alternately, [3-D Glasses Operation](#) shows how DLP Link pulses can be used instead.

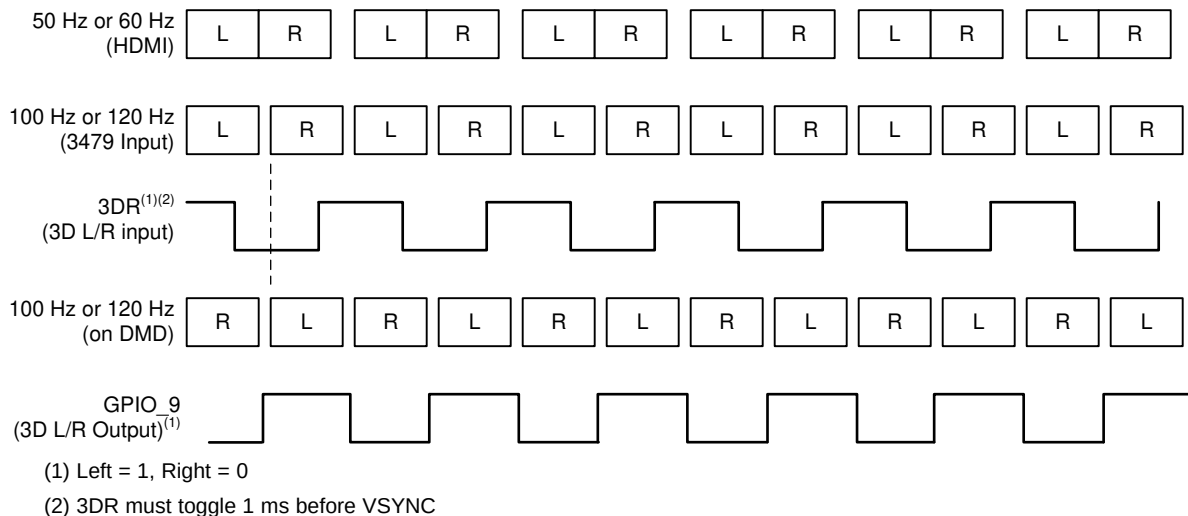


Figure 11. DLPC3479 L/R Frame and Signals Timing

7.2.2 Parallel Interface Supports Six Data Transfer Formats

The six data transfer formats supported by the parallel interface are:

- 24-bit RGB888 or 24-bit YCrCb888 on a 24 data wire interface
- 18-bit RGB666 or 18-bit YCrCb666 on a 18 data wire interface
- 16-bit RGB565 or 16-bit YCrCb565 on a 16 data wire interface

Input Source (continued)

- 16-bit YCrCb 4:2:2 (standard sampling assumed to be Y0Cb0, Y1Cr0, Y2Cb2, Y3Cr2, Y4Cb4, Y5Cr4, ...)

PDATA Bus – Parallel Interface Bit Mapping Modes shows the required PDATA(23:0) bus mapping for these six data transfer formats.

7.2.2.1 PDATA Bus – Parallel Interface Bit Mapping Modes

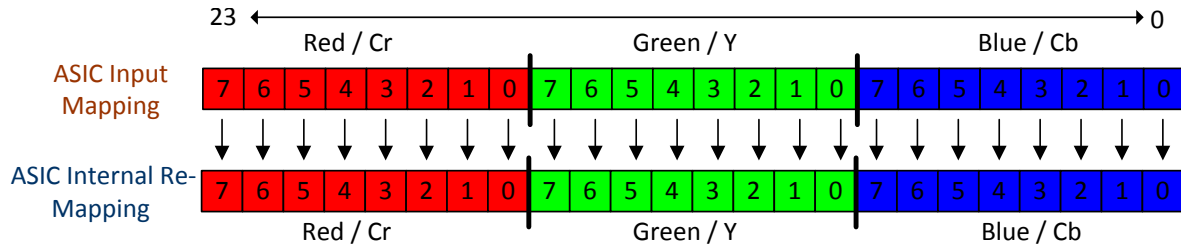


Figure 12. RGB-888 / YCrCb-888 I/O Mapping

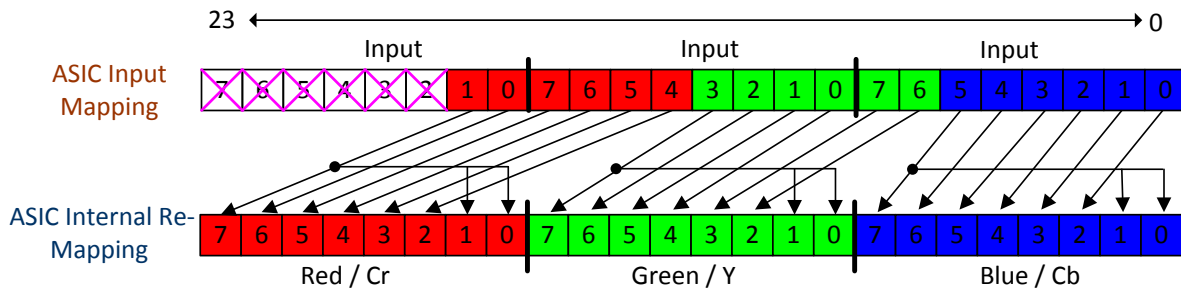


Figure 13. RGB-666 / YCrCb-666 I/O Mapping

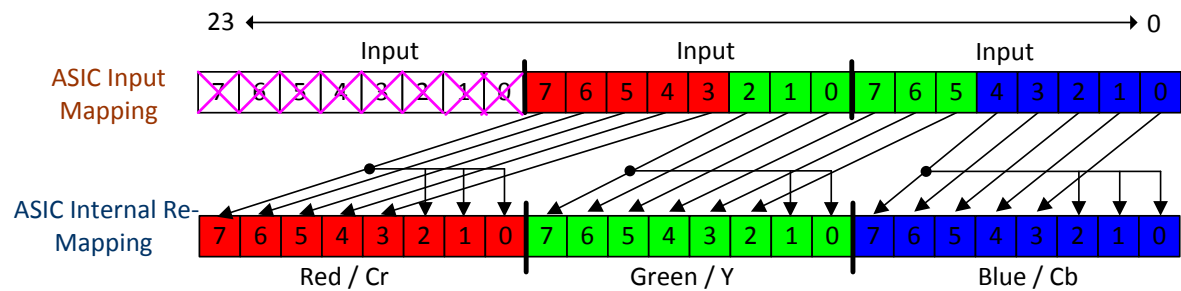


Figure 14. RGB-565 / YCrCb-565 I/O Mapping

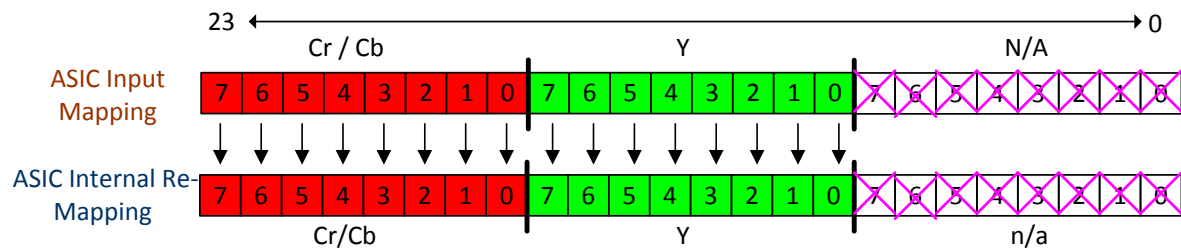


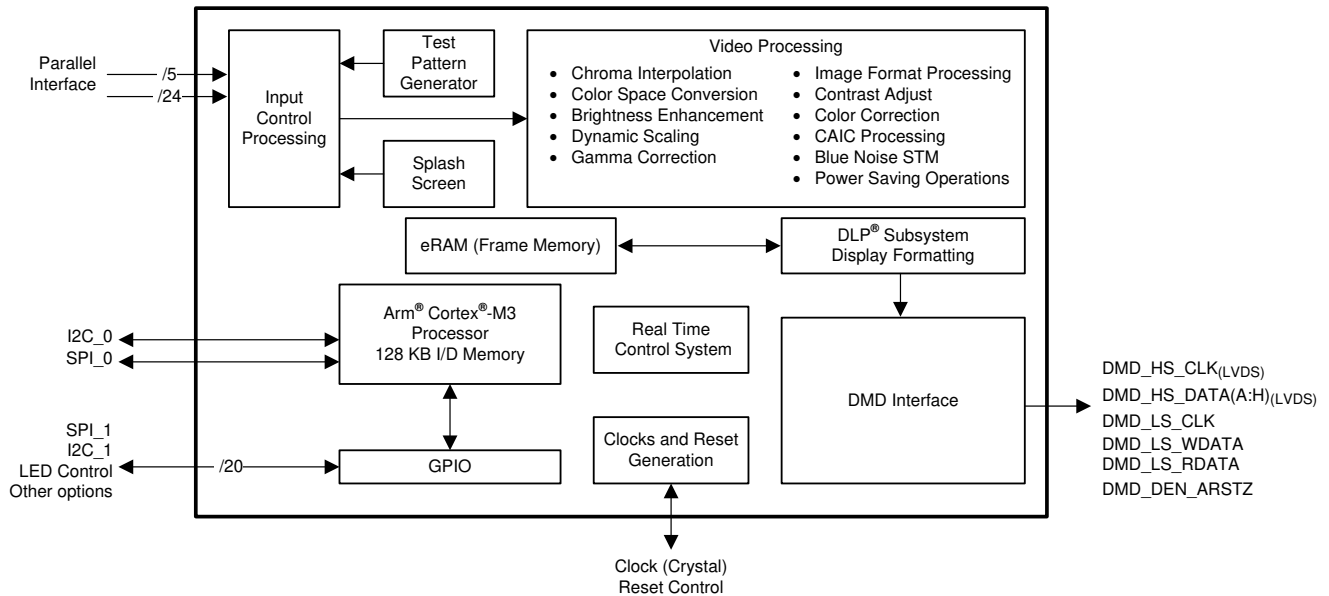
Figure 15. 16-Bit YCrCb-880 I/O Mapping

8 Detailed Description

8.1 Overview

The DLPC3479 is the display controller for the DLP4710 (.47 1080p) DMD. The DLPC3479 is part of the chipset comprising of two DLPC3479 controllers, the DLP4710 (.47 1080p) DMD, and the DLPA3000 or DLPA3005 PMIC/LED driver. All four components of the chipset must be used in conjunction with each other for reliable operation of the DLP4710 (.47 1080p) DMD. The 2xDLPC3479 controller provides a convenient interface between user electronics and the DMD to display data and steer light patterns with high speed, precision, and efficiency.

8.2 Functional Block Diagram



8.3 Feature Description

8.3.1 Pattern Display

Pattern display is one of the key capabilities of the DLPC3479 display and light controller. When the DLPC3479 controller is configured for pattern display, video processing functions can be bypassed to allow for accurate pattern display. For user flexibility and simple system design the DLPC3479 controller supports both external pattern and internal pattern streaming modes. In external pattern streaming mode, patterns are sent to the DLPC3479 controller over parallel interface. In internal pattern streaming mode, 1D patterns are pre-loaded in flash memory and a host command is sent to DLPC3479 controller to display the patterns. Internal pattern mode allows for a simple system design by eliminating the need for any external processor to generate and sent 1D patterns to the DLPC3479 controller.

The DLPC3479 controller outputs two configurable Trigger Out and one Trigger In signal to synchronize patterns with a camera, sensor, or other peripherals.

Table 3. Pattern Display Signals

SIGNAL NAME	DESCRIPTION
TRIG_OUT_1	External Pattern Mode: Active during each input frame.
	Internal Pattern Mode: Active during a predefined group of patterns.
TRIG_OUT_2	Active during display of each pattern. When operating in external pattern mode, one input frame can have multiple patterns.
TRIG_IN	Active in Internal Pattern Display mode only. Trigger In signal is used to advance to next patterns in internal pattern mode.

8.3.1.1 External Pattern Mode

External pattern mode supports 8-bit and 1-bit monochrome or RGB patterns.

8.3.1.1.1 8-bit Monochrome Patterns

In 8-bit external pattern mode, the DLPC3479 controller supports up to 120-Hz input frame rate (VSYNC). In this mode, the 24-bit input data sent over the parallel interface can be configured as a combination of 1 (8-bits), 2 (16-bits), or 3 (24-bits) 8-bit patterns. Equation 1 calculates the maximum pattern rate for 8-bit pattern.

$$120 \text{ Hz} \times 3 = 360 \text{ Hz}$$

where

- the maximum allowed input frame rate is 120 Hz (1)

The DLPC3479 controller firmware allows for the following user programmability.

- Exposure time (t_{Exposure}): Time during which a pattern displayed and the illumination is ON.
- DarkPre time (t_{DarkPre}): Dark time (before the pattern exposure) during which no pattern displays and the illumination is OFF.
- DarkPost time (t_{DarkPost}): Dark time (after the pattern exposure) during which no pattern displays and the illumination is OFF.
- Number of 8-bit patterns within a frame: 1, 2, or 3 within each Frame period
- Selection of Illuminator that is ON for each 8-bit pattern.
- TRIG_OUT_1 and TRIG_OUT_2 signal configuration and delay

Figure 16 shows a configuration with 3×8 -bit patterns.

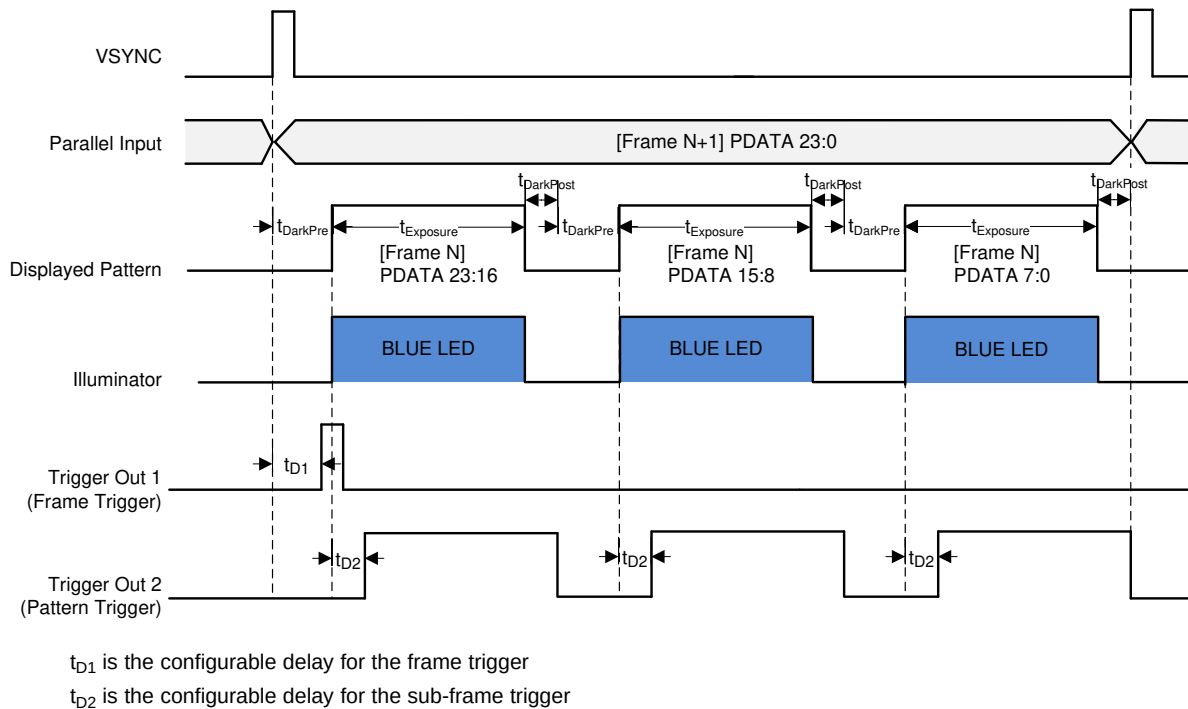


Figure 16. 3×8 -bit (Blue) Pattern Configurations

- 3×8 -bit patterns are displayed within each input VSYNC frame period.
- t_{DarkPre} , t_{Exposure} and t_{DarkPost} are the same for each pattern within a frame period.
- The sum of dark time and exposure time ($t_{\text{DarkPre}} + t_{\text{Exposure}} + t_{\text{DarkPost}}$) for the three patterns must be equal to or less than the full frame period. If the sum is less than the full frame period, additional dark time will be appended to the end of the last pattern.
- Blue LED is configured to be ON for each pattern.
- TRIG_OUT_1 (Frame Trigger) is configured active high polarity and will have a minimum pulse width of 20 microseconds. TRIG_OUT_1 delay (t_{D1}) is configured with respect to input VSYNC.

- TRIG_OUT_2 (Pattern Trigger) is configured active high polarity and stays active during the pattern exposure. TRIG_OUT_2 delay (t_{D2}) is configured with reference to the start of the pattern and is set once per pattern within a frame.

Figure 17 shows a configuration with 2×8 -bit patterns.

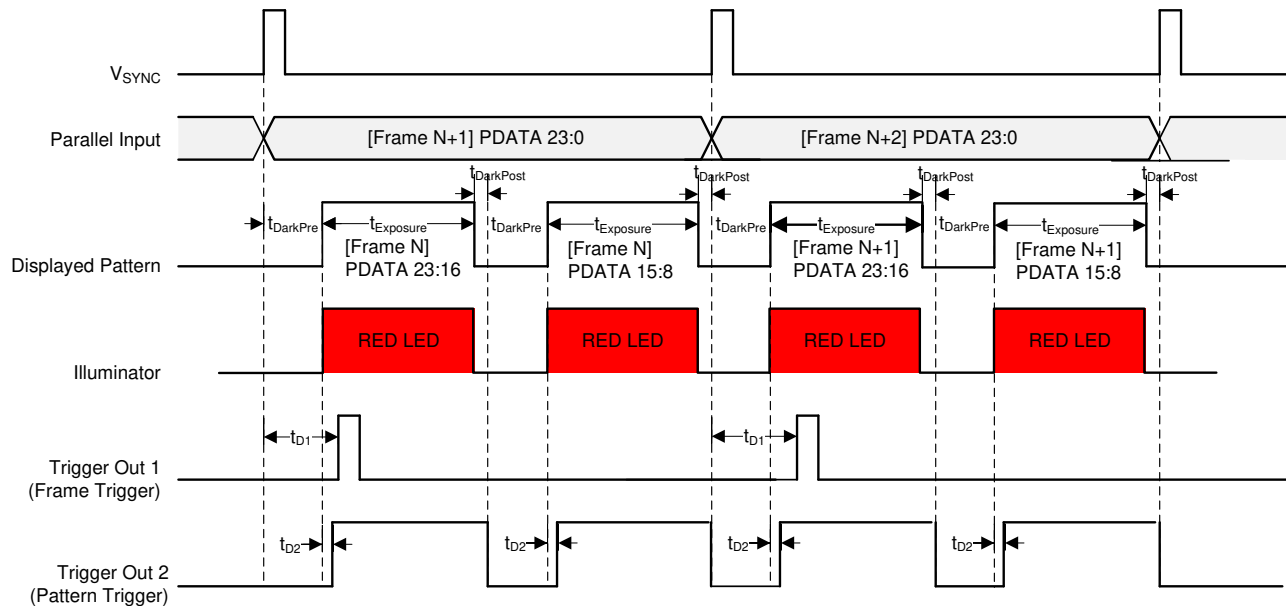


Figure 17. 2×8 -bit (Red) Pattern Configurations

- 2×8 -bit patterns are displayed within each input VSYNC frame period.
- $t_{DarkPre}$, $t_{Exposure}$ and $t_{DarkPost}$ are the same for each pattern within a frame period.
- The sum of dark time and exposure time ($t_{DarkPre} + t_{Exposure} + t_{DarkPost}$) for the three patterns must be equal to or less than the full frame period. If the sum is less than the full frame period, additional dark time will be appended to the end of the last pattern.
- Red LED is configured to be ON for each pattern.
- TRIG_OUT_1 (Frame Trigger) is configured active high polarity and will have a minimum pulse width of 20 microseconds. TRIG_OUT_1 delay (t_{D1}) is configured with respect to input V_{SYNC} .
- TRIG_OUT_2 (Pattern Trigger) is configured active high polarity and stays active during the pattern exposure. TRIG_OUT_2 delay (t_{D2}) is configured with reference to the start of the pattern and is set once per pattern within a frame.

Figure 18 shows a configuration with 1×8 -bit patterns.

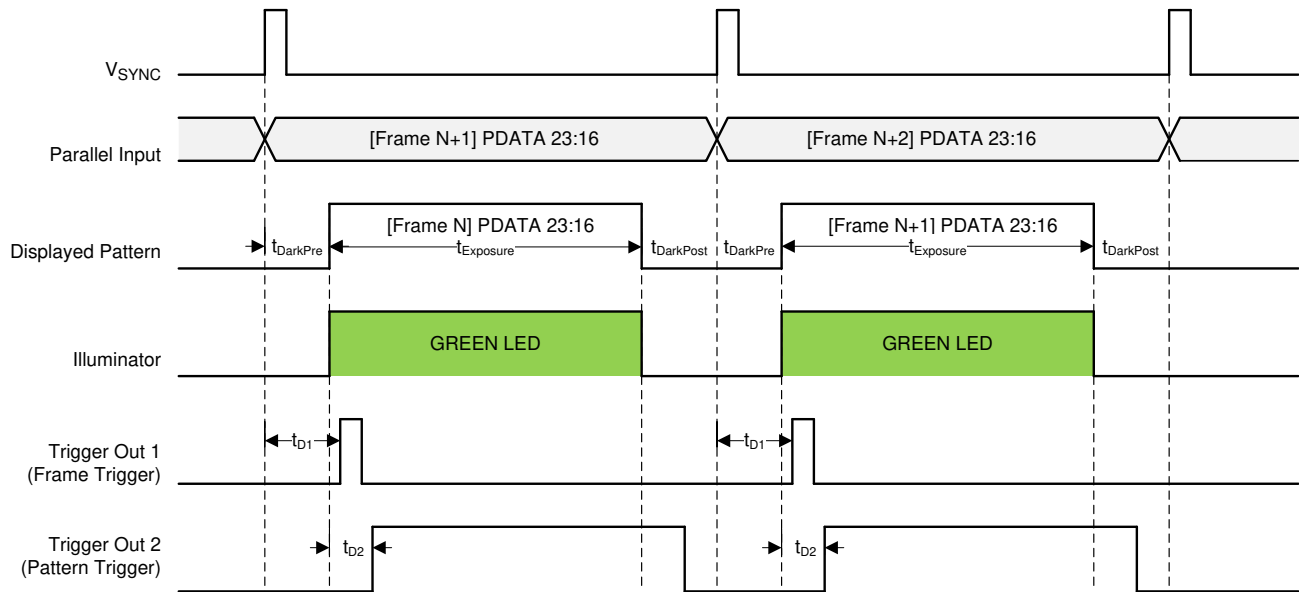


Figure 18. 1 × 8-bit (Green) Pattern Configurations

- 1 × 8-bit pattern is displayed within each input VSYNC frame period.
- t_{DarkPre} , t_{Exposure} and t_{DarkPost} are the same for each pattern within a frame period.
- The sum of dark time and exposure time ($t_{\text{DarkPre}} + t_{\text{Exposure}} + t_{\text{DarkPost}}$) for the three patterns must be equal to or less than the full frame period. If the sum is less than the full frame period, additional dark time will be appended to the end of the last pattern.
- Green LED is configured to be ON for each pattern.
- TRIG_OUT_1 (Frame Trigger) is configured active high polarity and will have a minimum pulse width of 20 microseconds. TRIG_OUT_1 delay (t_{D1}) is configured with respect to input V_{SYNC}.
- TRIG_OUT_2 (Pattern Trigger) is configured active high polarity and stays active during the pattern exposure. TRIG_OUT_2 delay (t_{D2}) is configured with reference to the start of the pattern and is set once per pattern within a frame.

8.3.1.1.2 1-Bit Monochrome Patterns

Similar to the 8-bit external pattern mode, the maximum supported input frame for 1-bit external pattern mode is 104.2 Hz. In 1-bit pattern mode each of the 24-bit inputs are treated as a separate binary patterns resulting in a maximum of 24 patterns. The maximum pattern rate for each 1-bit pattern is 2500 Hz.

The DLPC3479 controller firmware allows for the following user programmability:

- Exposure time: Time during which a pattern is displayed.
- Dark time: Time during which no pattern is displayed and the illumination is OFF.
- Number of 1-bit patterns within a frame- Up to maximum of 24.
- Illuminator: Illuminator that is ON for each 1-bit pattern. User defined illuminator is auto selected for all the patterns within a frame. User cannot select different illuminator for different 1-bit patterns within a frame.
- TRIG_OUT_1 and TRIG_OUT_2 signal configuration and delay.

Figure 19 shows a configuration with 24 × 1-bit patterns.

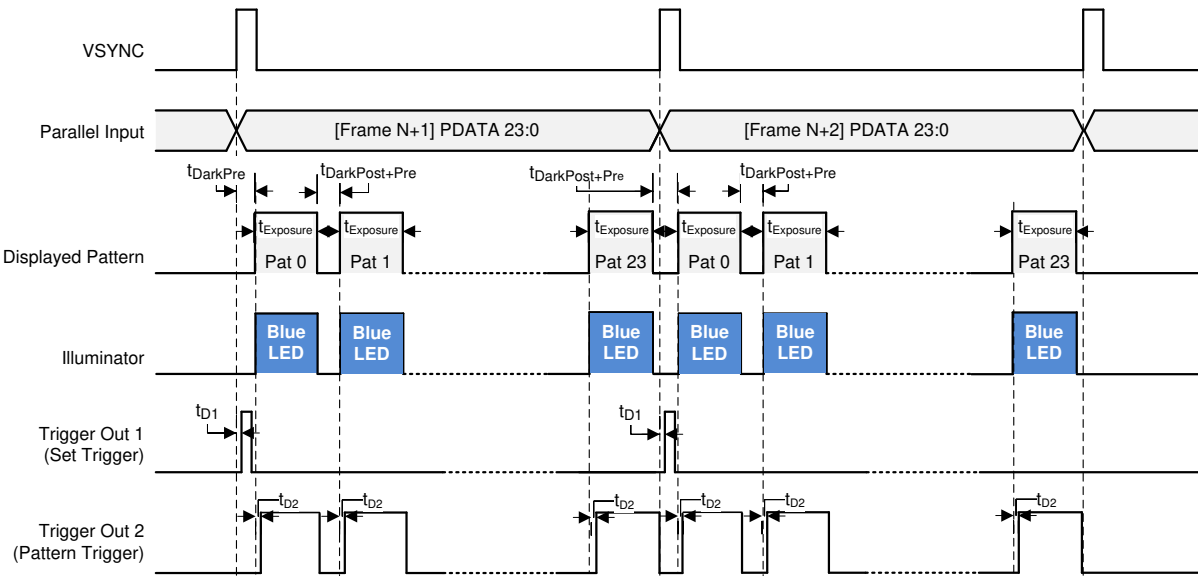


Figure 19. 24 × 1-bit (Blue) Pattern Configurations

- 24 × 1-bit patterns are displayed within each input VSYNC frame period.
- t_{DarkPre} , t_{Exposure} and t_{DarkPost} are the same for each pattern within a frame period.
- The sum of dark time and exposure time ($t_{\text{DarkPre}} + t_{\text{Exposure}} + t_{\text{DarkPost}}$) for the three patterns must be equal to or less than the full frame period. If the sum is less than the full frame period, additional dark time will be appended to the end of the last pattern.
- Blue LED is configured to be ON for each pattern.
- TRIG_OUT_1 (Frame Trigger) is configured active high polarity and will have a minimum pulse width of 20 microseconds. TRIG_OUT_1 delay (t_{D1}) is configured with respect to input V_{SYNC} .
- TRIG_OUT_2 (Pattern Trigger) is configured active high polarity and stays active during the pattern exposure. TRIG_OUT_2 delay (t_{D2}) is configured with reference to the start of the pattern and is set once per pattern within a frame.

8.3.1.2 Internal Pattern Mode

There are two key differences between internal and external pattern mode:

- Internal pattern mode only supports 1D patterns i.e the pattern data is same across the entire row or column of the DMD (Figure 20, Figure 21).
- Internal pattern mode enables user to design a simple system by eliminating need of an external processor to generate and send patterns every frame. In internal pattern mode one row or one column patterns are pre-loaded in the flash memory and a command is send to DLPC3479 controller to display the patterns. Implementation details on how to create patterns, save patterns in Flash memory and load patterns from flash memory into the DLPC3479 controller's internal memory are described in SW Programmers Guide.

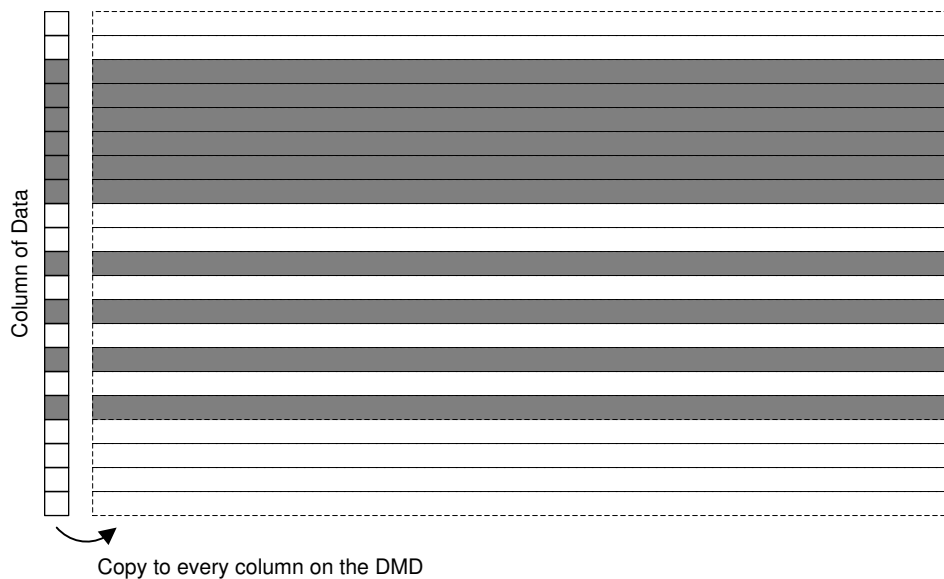


Figure 20. Column Replication

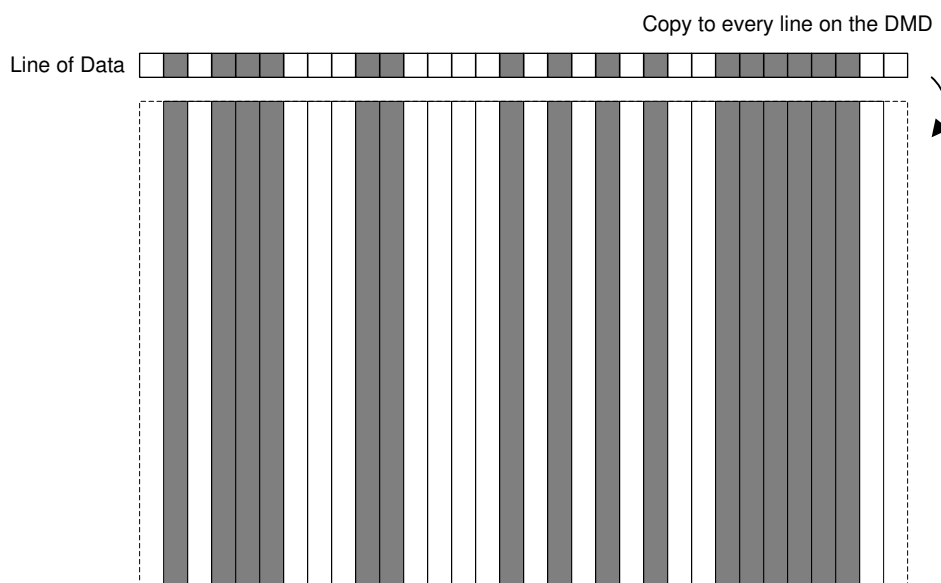


Figure 21. Row Replication

Internal pattern mode further provides two configurations to trigger the display of patterns, free running mode, (shown in [Figure 22](#)) and trigger in mode (shown in [Figure 23](#)).

8.3.1.2.1 Free Running Mode

In free running mode the DLPC3479 controller generates an internal synchronization signal to display pre-stored patterns. User sends an I²C command to instruct DLPC3479 controller to start download of the 1D patterns from flash memory into DLPC3479 controller's internal memory and displaying of the 1D patterns.

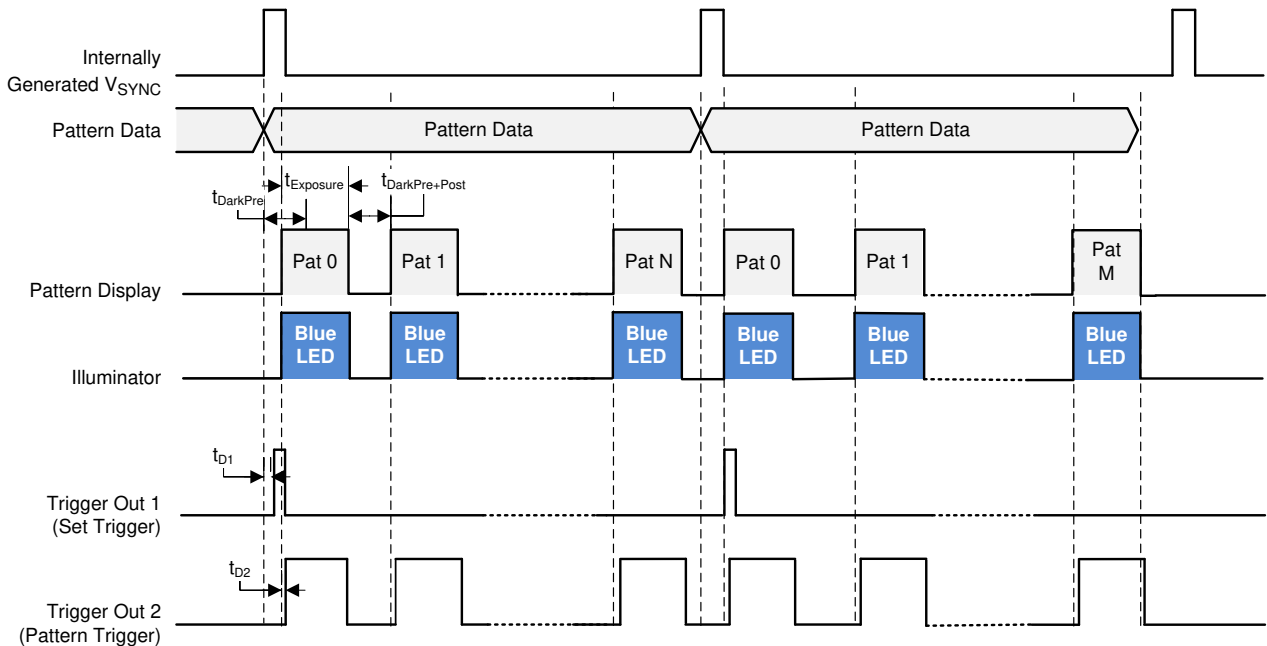
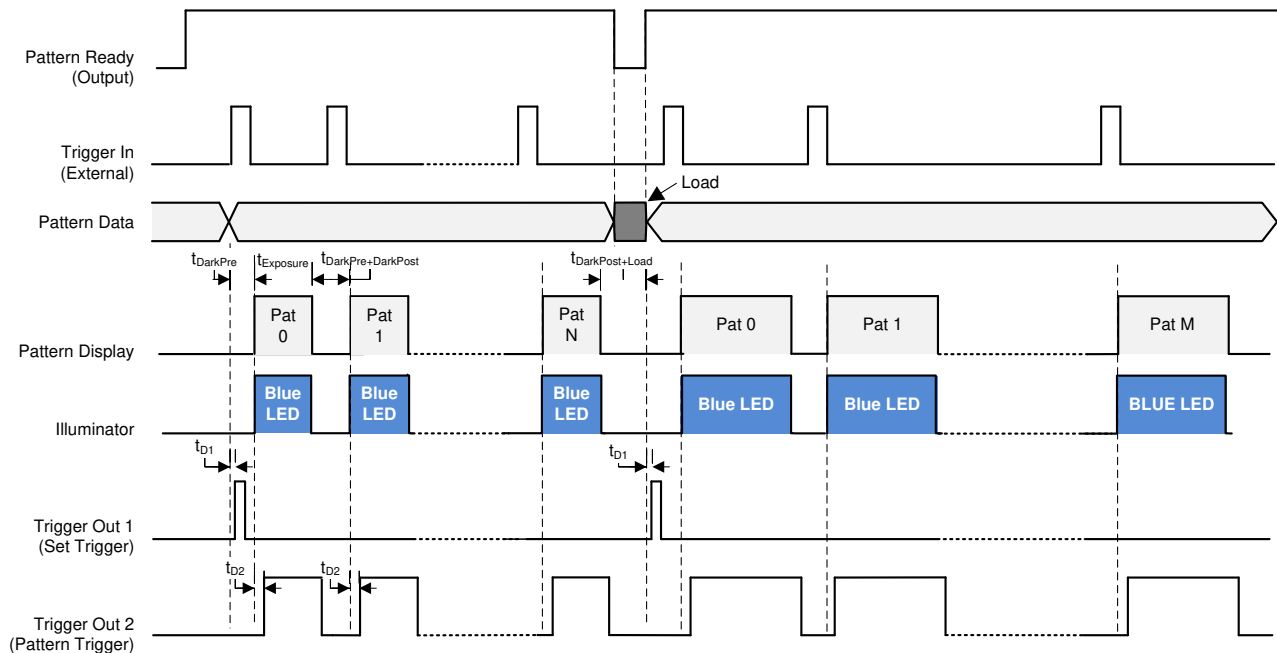


Figure 22. Free Running Mode

- The device displays multiple 1D patterns within an internally-generated V_{SYNC} signal. $t_{Exposure}$ (exposure time), $t_{DarkPre}$ and $t_{DarkPost}$ (dark time) are equal for all the 1D patterns within one internally generated V_{SYNC} frame.
- Blue LED is configured to be ON for each pattern.
- TRIG_OUT_1 (Frame Trigger) is configured active high polarity and will have a minimum pulse width of 20 microseconds. TRIG_OUT_1 delay (t_{D1}) is configured with respect to internally generated V_{SYNC} .
- TRIG_OUT_2 (Pattern Trigger) is configured active high polarity and stays active during the pattern exposure. TRIG_OUT_2 delay (t_{D2}) is configured with reference to the start of each pattern.
- V_{SYNC} is generated internally according to different sets of patterns stored in the SPI flash memory.

8.3.1.2.2 Trigger In Mode

Trigger In mode provides higher level of control to the user for displaying patterns. In this mode, user controls when to display the pattern by sending an external trigger signal to the DLPC3479 controller. The DLPC3479 controller outputs a Pattern Ready signal to let the user know when DLPC3479 controller is ready to accept the external trigger signal.


Figure 23. Trigger In Mode

- DLPC3479 controller sets the Pattern Ready signal high to denote that the DLPC3479 controller is ready to accept Trigger In signal.
- The user sends the external trigger input signal to the DLPC3479 controller to begin the display of the next pattern with t_{Exposure} (exposure time), t_{DarkPre} and t_{DarkPost} (dark time).
- Blue LED is configured to be ON for each pattern.
- TRIG_OUT_1 (Pattern Set Trigger) is configured active high polarity and will have a minimum pulse width of 20 microseconds. TRIG_OUT_1 delay (t_{D1}) is configured with respect to external trigger input (TRIG_IN).
- TRIG_OUT_2 (Pattern Trigger) is configured active high polarity and stays active during the pattern exposure. TRIG_OUT_2 delay (t_{D2}) is configured with reference to the start of each pattern exposure.

8.3.2 Interface Timing Requirements

This section defines the timing requirements for the external interfaces for the DLPC3479 ASIC.

8.3.2.1 Parallel Interface

The parallel interface complies with the standard graphics interface protocol, which includes a vertical synchronization signal (VSYNC_WE), horizontal synchronization signal (HSYNC_CS), optional data valid signal (DATAEN_CMD), a 24-bit data bus (PDATA), and a pixel clock (PCLK). The polarity of both syncs and the active edge of the clock are programmable. Figure 7 shows the relationship of these signals. The data valid signal (DATAEN_CMD) is optional in that the DLPC3479 provides auto-framing parameters that can be programmed to define the data valid window based on pixel and line counting relative to the horizontal and vertical syncs.

In addition to these standard signals, an optional side-band signal (PDM_CVS_TE) is available, which allows periodic frame updates to be stopped without losing the displayed image. When PDM_CVS_TE is active, it acts as a data mask and does not allow the source image to be propagated to the display. A programmable PDM polarity parameter determines if it is active high or active low. This parameter defaults to make PDM_CVS_TE active high; if this function is not desired, then tie the pin to a logic low on the PCB. PDM_CVS_TE is restricted to change only during vertical blanking.

NOTE

VSYSN_{WE} must remain active at all times (in lock-to-VSYNC mode) or the display sequencer will stop and cause the LEDs to be turned off.

8.3.3 Serial Flash Interface

The DLPC3479 uses an external SPI serial flash memory device for configuration support. The minimum required size is dependent on the desired minimum number of sequences, CMT tables, and splash options while the maximum supported size is 64 Mb.

For access to flash, the DLPC3479 uses a single SPI interface operating at a programmable frequency complying to industry standard SPI flash protocol. The programmable SPI frequency is defined to be equal to 180 MHz/N, where N is a programmable value between 5 to 127 providing a range from 36.0 to 1.41732 MHz. Note that this results in a relatively large frequency step size in the upper range (for example, 36 MHz, 30 MHz, 25.7 MHz, 22.5 MHz, and so forth) and thus this must be taken into account when choosing a flash device.

The DLPC3479 supports two independent SPI chip selects; however, the flash must be connected to SPI chip select zero (SPI0_CSZ0) because the boot routine is only executed from the device connected to chip select zero (SPI0_CSZ0). The boot routine uploads program code from flash to program memory, then transfers control to an auto-initialization routine within program memory. The DLPC3479 asserts the HOST_IRQ output signal high while auto-initialization is in progress, then drives it low to signal its completion to the host processor. Only after auto-initialization is complete will the DLPC3479 be ready to receive commands through I²C.

The DLPC3479 supports flash devices that are compatible with the modes of operation, features, and performance as defined in [Table 4](#) and [Table 5](#).

Table 4. SPI Flash Required Features or Modes of Operation

Feature	DLPC3479 Requirement
SPI interface width	Single
SPI protocol	SPI mode 0
Fast READ addressing	Auto-incrementing
Programming mode	Page mode
Page size	256 B
Sector size	4 kB sector
Block size	Any size
Block protection bits	0 = Disabled
Status register bit(0)	Write in progress (WIP, also called flash busy)
Status register bit(1)	Write enable latch (WEN)
Status register bits(6:2)	A value of 0 disables programming protection
Status register bit(7)	Status register write protect (SRWP)
Status register bits(15:8) (that is expansion status byte)	The DLPC3479 only supports single-byte status register R/W command execution, and thus may not be compatible with flash devices that contain an expansion status byte. However, as long as the expansion status byte is considered optional in the byte 3 position and any write protection control in this expansion status byte defaults to unprotected, then the device is compatible with the DLPC3479.

To support flash devices with program protection defaults of either enabled or disabled, the DLPC3479 always assumes the device default is enabled and goes through the process of disabling protection as part of the boot-up process. This process consists of:

- A write enable (WREN) instruction executed to request write enable, followed by
- A read status register (RDSR) instruction is then executed (repeatedly as needed) to poll the write enable latch (WEL) bit
- After the write enable latch (WEL) bit is set, a write status register (WRSR) instruction is executed that writes 0 to all 8-bits (this disables all programming protection)

Prior to each program or erase instruction, the DLPC3479 issues:

- A write enable (WREN) instruction to request write enable, followed by
- A read status register (RDSR) instruction (repeated as needed) to poll the write enable latch (WEL) bit

- After the write enable latch (WEL) bit is set, the program or erase instruction is executed
- Note the flash automatically clears the write enable status after each program and erase instruction

The specific instruction OpCode and timing compatibility requirements are listed in [Table 5](#) and [Table 6](#). However, the DLPC3479 device does not read the flash electronic signature ID and thus cannot automatically adapt protocol and clock rate based on the ID.

Table 5. SPI Flash Instruction OpCode and Access Profile Compatibility Requirements

SPI Flash Command	First Byte (OPCODE)	Second Byte	Third Byte	Fourth Byte	Fifth Byte	Sixth Byte
Fast READ (1 Output)	0x0B	ADDRS(0)	ADDRS(1)	ADDRS(2)	dummy	DATA(0) ⁽¹⁾
Read status	0x05	n/a	n/a	STATUS(0)		
Write status	0x01	STATUS(0)	⁽²⁾			
Write enable	0x06					
Page program	0x02	ADDRS(0)	ADDRS(1)	ADDRS(2)	DATA(0) ⁽¹⁾	
Sector erase (4KB)	0x20	ADDRS(0)	ADDRS(1)	ADDRS(2)		
Chip erase	0xC7					

(1) Only the first data byte is shown, data continues.

(2) DLPC3479 does not support access to a second/expansion Write Status byte.

The specific and timing compatibility requirements for a DLPC3479 compatible flash are listed in [Table 6](#) and [Table 7](#).

Table 6. SPI Flash Key Timing Parameter Compatibility Requirements⁽¹⁾⁽²⁾

SYMBOL	ALTERNATE SYMBOL		MIN	MAX	UNIT
FR	f_c	Access frequency (all commands)	≤1.42		MHz
t_{SHSL}	t_{CSH}	Chip select high time (also called chip select deselect time)	≤200		ns
t_{CLQX}	t_{HO}	Output hold time	≥0		ns
t_{CLQV}	t_v	Clock low to output valid time		≤11	ns
t_{DVCH}	t_{DSU}	Data in set-up time	≤5		ns
t_{CHDX}	t_{DH}	Data in hold time	≤5		ns

(1) The timing values are related to the specification of the flash device itself, not the DLPC3479.

(2) The DLPC3479 does not drive the HOLD or WP (active low write protect) pins on the flash device, and thus tie these pins to a logic high on the PCB through an external pullup.

The DLPC3479 supports 1.8-, 2.5-, or 3.3-V serial flash devices. To do so, V_{CC_FLSH} must be supplied with the corresponding voltage. [Table 7](#) contains a list of 1.8-, 2.5-, and 3.3-V compatible SPI serial flash devices supported by the DLPC3479.

8.3.4 Tested Flash Devices

Table 7. DLPC3479 Compatible SPI Flash Device Options (3.3-V Compatible Devices)^{(1) (2)}

DVT ⁽³⁾	DENSITY (Mb)	VENDOR	PART NUMBER	PACKAGE SIZE
Yes	32 Mb	Winbond	W25Q32FVSSIG	5.2 mm x 7.9 mm, 8-pin SOIC
Yes	64 Mb	Winbond	W25Q64FVSSIG	5.2 mm x 7.9 mm, 8-pin SOIC
Yes	32 Mb	Macronix	MX25L3233FM2I-08G	5.2 mm x 7.9 mm, 8-pin SOP
Yes	64 Mb	Macronix	MX25L6433FM2I-08G	5.2 mm x 7.9 mm, 8-pin SOP

- (1) The flash supply voltage must match V_{CC_FLSH} on the DLPC3479. Special attention is needed when ordering devices to ensure the desired supply voltage is attained as multiple voltage options are often available under the same base part number.
- (2) Beware when considering Numonyx (Micron) serial flash devices as they typically do not have the 4 kB sector size needed to be DLPC3479 compatible.
- (3) All of the flash devices shown are compatible with the DLPC3479, but only those marked with 'Yes' in the DVT column have been validated during TI Validation testing using a TI reference design. Those marked with 'No' can be used at the original design manufacturer (ODM)'s own risk. Other parts aside from those shown can be used if the timing conditions in [Serial Flash Interface](#) are met.

8.3.5 Serial Flash Programming

Note that the flash can be programmed through the DLPC3479 over I²C or by driving the SPI pins of the flash directly while the DLPC3479 I/O are tri-stated. SPI0_CLK, SPI0_DOUT, and SPI0_CSZ0 I/O can be tri-stated by holding RESETZ in a logic low state while power is applied to the DLPC3479. Note that SPI0_CSZ1 is not tri-stated by this same action.

8.3.6 SPI Signal Routing

The DLPC3479 is designed to support two SPI slave devices on the SPI0 interface, specifically a serial flash and the DLPA3000 or DLPA3005. This requires routing associated SPI signals to two locations while attempting to operate up to 36 MHz. Take special care to ensure that reflections do not compromise signal integrity. To this end, the following recommendations are provided:

- Split the SPI0_CLK PCB signal trace from the DLPC3479 source to each slave device into separate routes as close to the DLPC3479 as possible. Make the SPI0_CLK trace length to each device equal in total length.
- Split the SPI0_DOUT PCB signal trace from the DLPC3479 source to each slave device into separate routes as close to the DLPC3479 as possible. Make the SPI0_DOUT trace length to each device equal in total length (use the same strategy as SPI0_CLK).
- Make the SPI0_DIN PCB signal trace from each slave device to the point where they intersect on their way back to the DLPC3479 equal in length and as short as possible. Make sure they share a common trace back to the DLPC3479.
- SPI0_CSZ0 and SPI0_CSZ1 need no special treatment because they are dedicated signals which drive only one device.

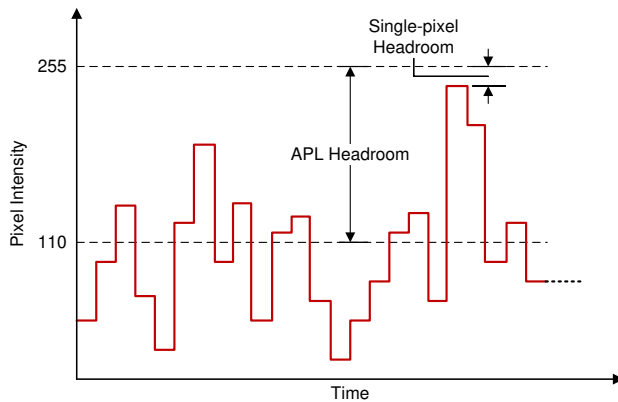
8.3.7 I²C Interface Performance

Both DLPC3479 I²C interface ports support a 100-kHz baud rate. By definition, I²C transactions operate at the speed of the slowest device on the bus, thus there is no requirement to match the speed grade of all devices in the system.

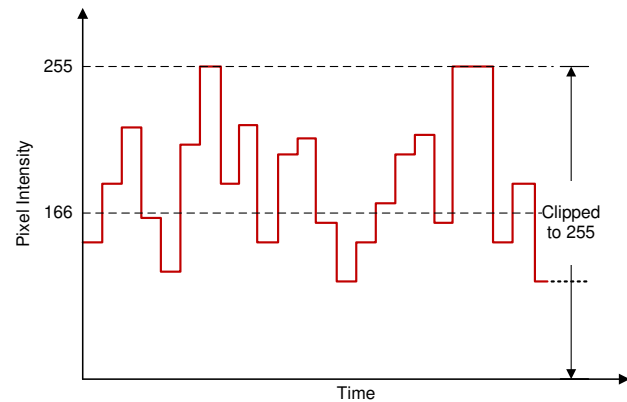
8.3.8 Content-Adaptive Illumination Control

Content Adaptive Illumination control (CAIC) is part of the IntelliBright™ suite of advanced image processing algorithms that adaptively enhances brightness and reduces power. In common real-world image content most pixels in the images are well below full scale for the R (red), G (green), and B (blue) digital channels input to the DLPC34xx. As a result of this, the average picture level (APL) for the overall image is also well below full scale, and the dynamic range for the collective set of pixel values is not fully used. CAIC takes advantage of the headroom between the source image APL and the top of the available dynamic range of the display system.

CAIC evaluates images on a frame-by-frame basis and derives three unique digital gains, one for each of the R, G, and B color channel. During image processing, CAIC applies each gain to all pixels in the associated color channel. The calculated gain is applied to all pixels in that channel so that the pixels as a group collectively shift upward and as close to full scale as possible. To prevent any image quality degradation, the gains are set at the point where just a few pixels in each color channel are clipped. Figure 24 and Figure 25 show an example of the application of CAIC for one color channel.



(1) APL = 110

Figure 24. Source Pixels for a Color Channel


(1) APL = 166

(2) Channel gain = $166/110 = 1.51$
Figure 25. Pixels for a Color Channel After CAIC Processing

Figure 25 shows the gain that is applied to a color processing channel inside the DLPC34xx. Additionally, CAIC adjusts the power for the R, G, and B LED by commanding different LED currents. For each color channel of an individual frame, CAIC intelligently determines the optimal combination of digital gain and LED power. The user configurable CAIC settings heavily influence the amount of digital gain that is applied to a color channel and the LED power for that color.

As CAIC applies a digital gain to each color channel and adjusts the power to each LED, CAIC ensures the resulting color balance in the final image matches the target color balance for the projector system. Thus, the effective displayed white point of images is held constant by CAIC from frame to frame.

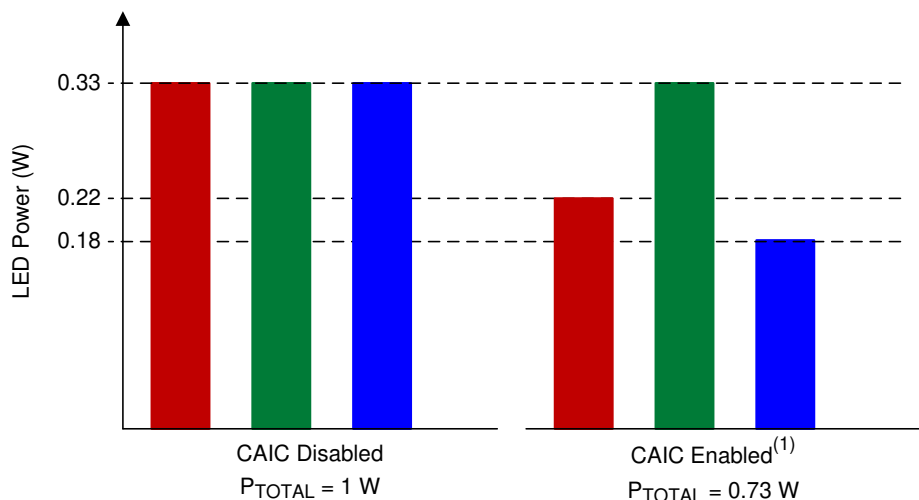
CAIC can be used to increase the overall image brightness while holding the total power for all LEDs constant, or CAIC can be used to hold the overall image brightness constant while decreasing LED power. In summary, CAIC has two primary modes of operation:

- Power reduction mode holds overall image brightness constant while reducing LED power
- Enhanced brightness mode holds overall LED power constant while enhancing image brightness

In power reduction mode, since the R, G, and B channels can be gained up by CAIC inside the DLPC34xx, the LED power can be reduced for any color channel until the brightness of the color on the screen is unchanged. Thus, CAIC can achieve an overall LED power reduction while maintaining the same overall image brightness as if CAIC was not used. Figure 26 shows an example of LED power reduction by CAIC for an image where the red and blue LEDs can consume less power.

In enhanced brightness mode the R, G, and B channels can be gained up by CAIC with LED power generally being held constant. This results in an enhanced brightness with no power savings.

While there are two primary modes of operation described, the DLPC34xx actually operates within the extremes of pure power reduction mode and enhanced brightness mode. The user can configure which operating mode the DLPC34xx will more closely follow by adjusting the CAIC gain setting as described in the software programmer's guide.



(1) With CAIC enabled, if red and blue LEDs require less than nominal power for a given input image, the red and blue LED power will reduce.

Figure 26. CAIC Power Reduction Mode (for Constant Brightness)

8.3.9 Local Area Brightness Boost (LABB)

LABB is an image processing algorithm that adaptively gains up regions of an image that are dim relative to the average picture level. Some regions of the image will have significant gain applied, and some regions will have little or no gain applied. LABB evaluates images frame by frame and derives the local area gains to be used uniquely for each image. Since many images have a net overall boost in gain even if some parts of the image receive no gain, the overall perceived brightness of the image is boosted.

Figure 27 shows a split screen example of the impact of the LABB algorithm for an image that includes dark areas.



图 27. Boosting Brightness in Local Areas of an Image

LABB works best when the decision about the strength of gains used is determined by ambient light conditions. For this reason, an ambient light sensor can be read by an external processor for each frame. Based on the sensor readings, the external processor can send LABB strength commands to apply higher gains for bright rooms to help overcome any washing out of images. LABB will receive commands to apply lower gains in dark rooms to prevent over-punching of images.

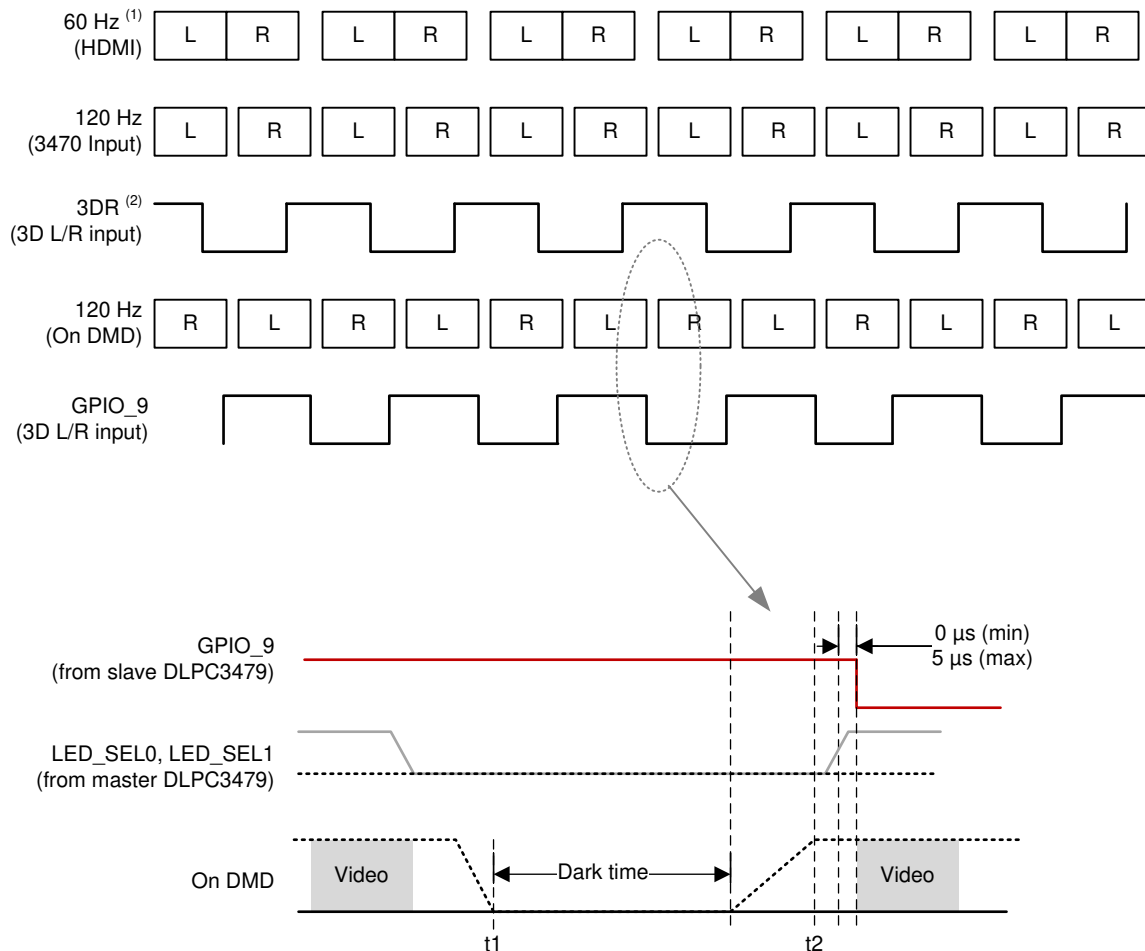
8.3.10 3-D Glasses Operation

For supporting 3D glasses, the DLPC3479-based chip set outputs synchronization information to synchronize the Left eye/Right eye shuttering in the glasses with the displayed DMD image frames.

Two different types of glasses are often used to achieve synchronization. One relies on an IR transmitter on the system PCB to send an IR synchronization signal to an IR receiver in the glasses. In this case, the DLPC3479 output signal GPIO_09 can be used to cause the IR transmitter to send an IR synchronization signal to the glasses. The timing for signal GPIO_09 is shown in Figure 11.

The second type of glasses relies on synchronization information that is encoded into the light being output from the projection lens. This is referred to as the DLP Link approach for 3D, and many 3D glasses from different suppliers have been built using this method. This demonstrates that the DLP Link method can work reliably. The advantage of the DLP Link approach is that it takes advantage of existing projector hardware to transmit the synchronization information to the glasses. This can save on cost, size, and power in the projector.

For generating the DLP Link synchronization information, one light pulse per DMD frame is output from the projection lens while the glasses have both shutters closed. To achieve this, the DLPC3479 will tell the DLPA3000 or DLPA3005 when to turn on the illumination source (typically LEDs or lasers) so that an encoded light pulse is output once per DMD frame. Since the shutters in the glasses are both off when the DLP Link pulse is sent, the projector illumination source will also be off except for the when light is sent to create the DLP Link pulse. The timing for the light pulses for DLP Link 3D operation is shown in Figure 28 and Figure 29.



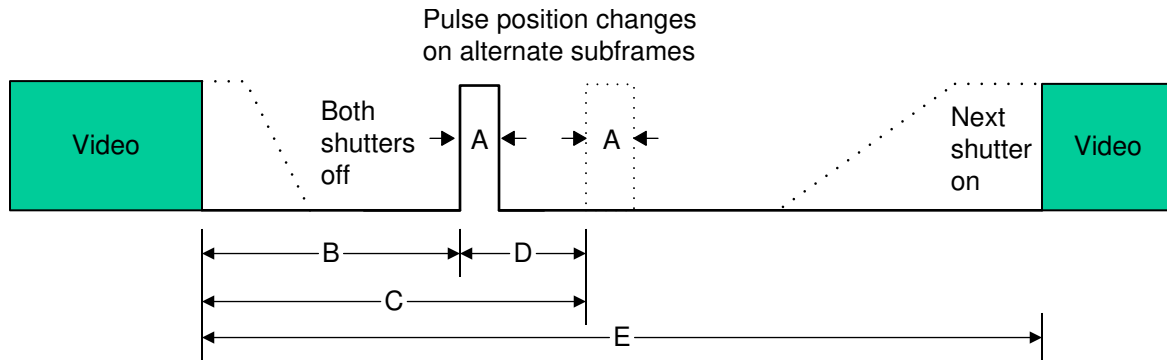
(1) Left = 1, Right = 0

(2) 3DR must toggle 1 ms before VSYNC

t1: both shutters off

t2: next shutter on

图 28. DLPC3479 L/R Timing for DLP Link



D: The period between DLPLink pulses alternates between the subframe period + D and the subframe period – D, where D is the delta period.

图 29. 3D DLP Link Pulse Timing

表 8. 3D DLP Link Nominal Timing Table⁽¹⁾

HDMI SOURCE REFERENCE (Hz)	3D DMD SEQUENCE RATE (Hz)	A (μs)	B (μs)	C (μs)	D (μs)	E (μs)
49.0	98	31.8	>500	>622	161.6	>2000
50.0	100	31.2	>500	>658	158.4	>2000
51.0	102	30.6	>500	>655	155.3	>2000
59.0	118	26.4	>500	>634	134.2	>2000
60.0	120	26.0	>500	>632	132.0	>2000
61.0	122	25.6	>500	>630	129.8	>2000

(1) Timing parameter C is always the sum of B+D.

表 9. 3D DLP Link Min/Max Timing Table⁽¹⁾

HDMI SOURCE REFERENCE (Hz)	3D DMD SEQUENCE RATE (Hz)	A (μs)	B (μs)	C (μs)	D (μs)	E (μs)
49.0	98	20-32	>500	>628	128-163	>2000
50.0	100	20-32	>500	>628	128-163	>2000
51.0	102	20-32	>500	>628	128-163	>2000
59.0	118	20-32	>500	>628	128-163	>2000
60.0	120	20-32	>500	>628	128-163	>2000
61.0	122	20-32	>500	>628	128-163	>2000

(1) Timing parameter C is always the sum of B+D.

8.3.11 DMD (Sub-LVDS) Interface

The DLPC3479 ASIC DMD interface consists of a HS 1.8-V sub-LVDS output only interface with a maximum clock speed of 600-MHz DDR and a LS SDR (1.8-V LVCMOS) interface with a fixed clock speed of 120 MHz. 表 10 shows how the 8 sub-LVDS lanes are configured for the DLP4710 (.47 1080p) DMD.

表 10. DLP4710 (.47 1080p) DMD – ASIC to 8-Lane DMD Pin Mapping

DLPC3479 ASIC 8 LANE DMD ROUTING OPTION #1		
MASTER DLPC3479 PINS	SLAVE DLPC3479 PINS	DMD PINS
HS_WDATA_D_P HS_WDATA_D_N	HS_WDATA_E_P HS_WDATA_E_N	Input DATA_p_0 Input DATA_n_0
HS_WDATA_C_P HS_WDATA_C_N	HS_WDATA_F_P HS_WDATA_F_N	Input DATA_p_1 Input DATA_n_1
HS_WDATA_B_P HS_WDATA_B_N	HS_WDATA_G_P HS_WDATA_G_N	Input DATA_p_2 Input DATA_n_2

表 10. DLP4710 (.47 1080p) DMD – ASIC to 8-Lane DMD Pin Mapping (接下页)

DLPC3479 ASIC 8 LANE DMD ROUTING OPTION #1		
HS_WDATA_A_P HS_WDATA_A_N	HS_WDATA_H_P HS_WDATA_H_N	Input DATA_p_3 Input DATA_n_3
HS_WDATA_H_P HS_WDATA_H_N	HS_WDATA_A_P HS_WDATA_A_N	Input DATA_p_4 Input DATA_n_4
HS_WDATA_G_P HS_WDATA_G_N	HS_WDATA_B_P HS_WDATA_B_N	Input DATA_p_5 Input DATA_n_5
HS_WDATA_F_P HS_WDATA_F_N	HS_WDATA_C_P HS_WDATA_C_N	Input DATA_p_6 Input DATA_n_6
HS_WDATA_E_P HS_WDATA_E_N	HS_WDATA_D_P HS_WDATA_D_N	Input DATA_p_7 Input DATA_n_7
DLPC3479 ASIC 8 LANE DMD ROUTING OPTION #2		
MASTER DLPC3479 PINS	SLAVE DLPC3479 PINS	DMD PINS
HS_WDATA_E_P HS_WDATA_E_N	HS_WDATA_D_P HS_WDATA_D_N	Input DATA_p_0 Input DATA_n_0
HS_WDATA_F_P HS_WDATA_F_N	HS_WDATA_C_P HS_WDATA_C_N	Input DATA_p_1 Input DATA_n_1
HS_WDATA_G_P HS_WDATA_G_N	HS_WDATA_B_P HS_WDATA_B_N	Input DATA_p_2 Input DATA_n_2
HS_WDATA_H_P HS_WDATA_H_N	HS_WDATA_A_P HS_WDATA_A_N	Input DATA_p_3 Input DATA_n_3
HS_WDATA_A_P HS_WDATA_A_N	HS_WDATA_H_P HS_WDATA_H_N	Input DATA_p_4 Input DATA_n_4
HS_WDATA_B_P HS_WDATA_B_N	HS_WDATA_G_P HS_WDATA_G_N	Input DATA_p_5 Input DATA_n_5
HS_WDATA_C_P HS_WDATA_C_N	HS_WDATA_F_P HS_WDATA_F_N	Input DATA_p_6 Input DATA_n_6
HS_WDATA_D_P HS_WDATA_D_N	HS_WDATA_E_P HS_WDATA_E_N	Input DATA_p_7 Input DATA_n_7

8.3.12 Calibration and Debug Support

The DLPC3479 contains a test point output port, TSTPT_(7:0), which provides selected system calibration support as well as ASIC debug support. These test points are inputs while reset is applied and switched to outputs when reset is released. The state of these signals is sampled upon the release of system reset and the captured value configures the test mode until the next time reset is applied. Each test point includes an internal pulldown resistor, thus external pullups must be used to modify the default test configuration. The default configuration (x000) corresponds to the TSTPT_(7:0) outputs remaining tri-stated to reduce switching activity during normal operation. For maximum flexibility, an option to jumper to an external pullup is recommended for TSTPT_(2:0). Pullups on TSTPT_(6:3) are used to configure the ASIC for a specific mode or option. TI does not recommend adding pullups to TSTPT_(7:3) because this has adverse effects for normal operation. This external pullup is only sampled upon a 0-to-1 transition on the RESETZ input, thus changing their configuration after reset is released will not have any effect until the next time reset is asserted and released. 表 11 defines the test mode selection for one programmable scenario defined by TSTPT(2:0).

表 11. Test Mode Selection Scenario Defined by TSTPT(2:0)⁽¹⁾

TSTPT(2:0) CAPTURE VALUE	NO SWITCHING ACTIVITY x000	CLOCK DEBUG OUTPUT x010
TSTPT(0)	HI-Z	60 MHz
TSTPT(1)	HI-Z	30 MHz
TSTPT(2)	HI-Z	0.7 to 22.5 MHz
TSTPT(3)	HI-Z	HIGH
TSTPT(4)	HI-Z	LOW
TSTPT(5)	HI-Z	HIGH

(1) These are only the default output selections. Software can reprogram the selection at any time.

表 11. Test Mode Selection Scenario Defined by TSTPT(2:0)⁰ (接下页)

TSTPT(2:0) CAPTURE VALUE	NO SWITCHING ACTIVITY x000	CLOCK DEBUG OUTPUT x010
TSTPT(6)	HI-Z	HIGH
TSTPT(7)	HI-Z	7.5 MHz

8.3.13 DMD Interface Considerations

The sub-LVDS HS interface waveform quality and timing on the DLPC3479 ASIC is dependent on the total length of the interconnect system, the spacing between traces, the characteristic impedance, etch losses, and how well matched the lengths are across the interface. Thus, ensuring positive timing margin requires attention to many factors.

As an example, DMD interface system timing margin can be calculated as follows:

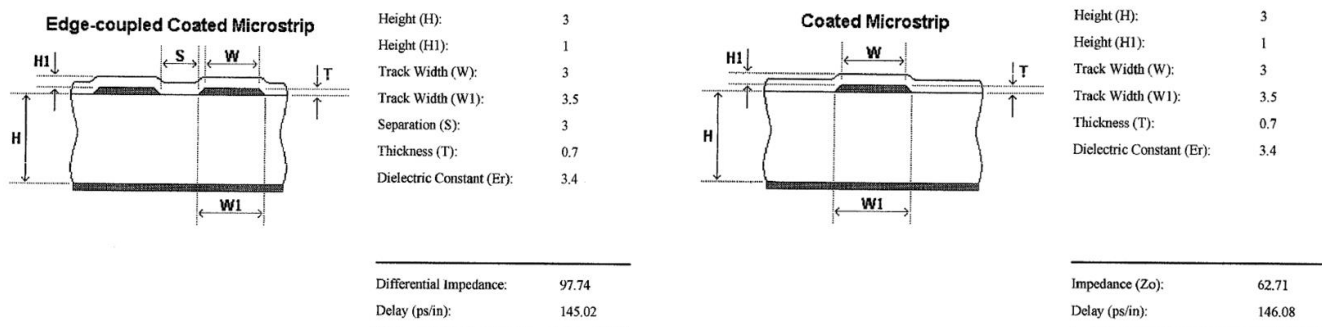
$$\text{Setup Margin} = (\text{DLPC3479 output setup}) - (\text{DMD input setup}) - (\text{PCB routing mismatch}) - (\text{PCB SI degradation}) \quad (2)$$

$$\text{Hold-time Margin} = (\text{DLPC3479 output hold}) - (\text{DMD input hold}) - (\text{PCB routing mismatch}) - (\text{PCB SI degradation})$$

Where PCB SI degradation is signal integrity degradation due to PCB effects which includes such things as Simultaneously Switching Output (SSO) noise, cross-talk and Inter-symbol Interference (ISI) noise. (3)

DLPC3479 I/O timing parameters as well as DMD I/O timing parameters can be found in their corresponding data sheets. Similarly, PCB routing mismatch can be budgeted and met through controlled PCB routing. However, PCB SI degradation is a more complicated adjustment.

In an attempt to minimize the signal integrity analysis that would otherwise be required, the following PCB design guidelines are provided as a reference of an interconnect system that will satisfy both waveform quality and timing requirements (accounting for both PCB routing mismatch and PCB SI degradation). Variation from these recommendations may also work, but should be confirmed with PCB signal integrity analysis or lab measurements.



LEFT: DMD_HS Differential Signals

RIGHT: DMD_LS Signals

图 30. DMD Interface Board Stack-Up Details

8.4 Device Functional Modes

The DLPC3479 has two functional modes (ON/OFF) controlled by a single pin PROJ_ON:

- When pin PROJ_ON is set high, the projector automatically powers up and an image is projected from the DMD.
- When pin PROJ_ON is set low, the projector automatically powers down and only microwatts of power are consumed.

9 Application and Implementation

注

Information in the following applications sections is not part of the TI component specification, and TI does not warrant its accuracy or completeness. TI's customers are responsible for determining suitability of components for their purposes. Customers should validate and test their design implementation to confirm system functionality.

9.1 Application Information

The DLPC3479 controller is required to be coupled with the DLP4710 (.47 1080p) DMD to provide a reliable display solution for various display and light control applications. DMDs are spatial light modulators which reflect incoming light from an illumination source to one of two directions, with the primary direction being into a projection or collection optic. Each application is derived primarily from the optical architecture of the system and the format of the data coming into the DLPC3479. Applications of interest include accessory projectors, wearable (near-eye or head mounted) displays, interactive display, low latency gaming display, digital signage, high resolution 3D printing products and high accuracy and small form factor 3D depth capture products. This section describes typical 3D depth capture DLP systems using internal pattern streaming modes. In internal patterns streaming mode, structured light patterns are stored in flash memory and directly displayed by the DLPC3479 controller without any need to stream the patterns over 24 bit parallel interface to DLPC3479.

9.2 Typical Application

DLPC3479 controller with DLP4710 DMD enables high accuracy and small form factor 3D depth capture products. This section shows a typical 3D depth capture system block diagram using internal pattern streaming mode.

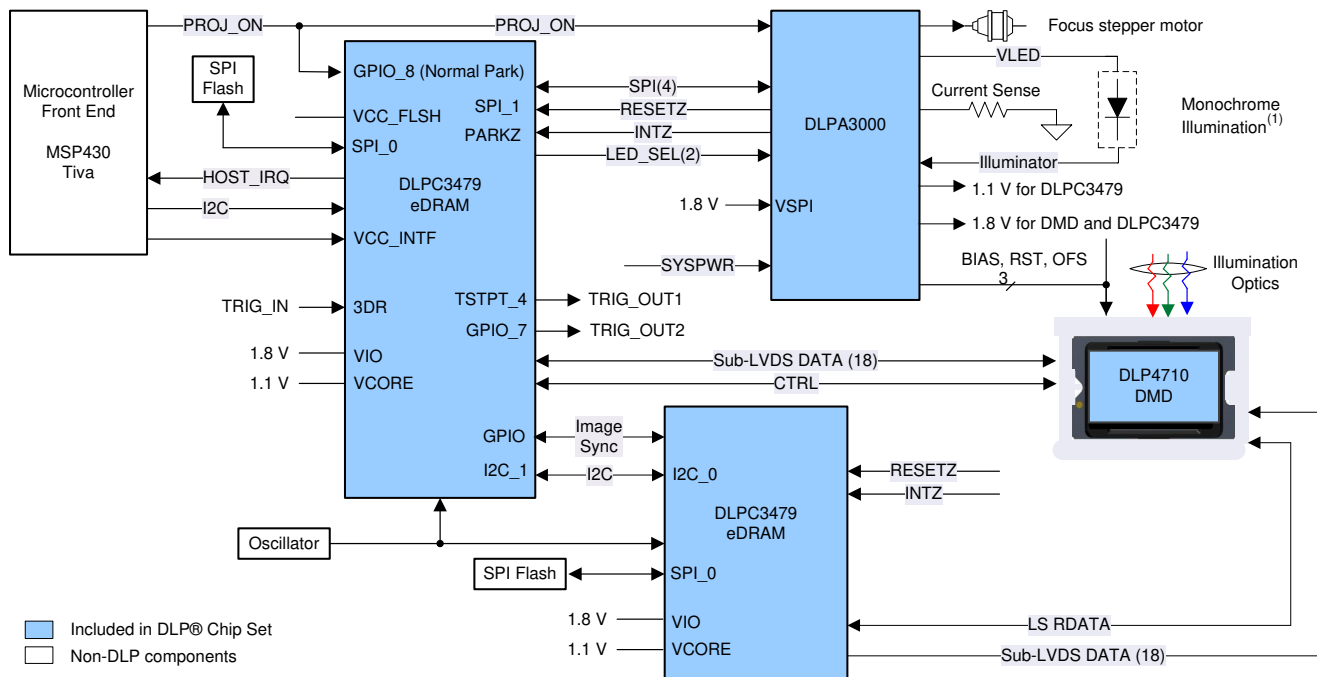


图 31. Typical Application Diagram

Typical Application (接下页)

9.2.1 Design Requirements

A high accuracy 3D depth capture product is created by using a DLP chipset comprised of DLP4710 DMD, 2xDLPC3479 controller and DLPA3000 PMIC/LED drive. The DLPC3479 simplifies the pattern generation, the DLPA300 provides the needed analog functions and DMD displays the required patterns for accurate 3D depth capture. In addition to the three DLP devices in the chipset, other components may be required to complete the application. Minimally, a flash component is required to store patterns, the software, and the firmware in order to control the DLPC3479 controller. DLPC3479 controller supports any illumination source including IR light source (LEDs or VCSEL), UV light source or visible light source (Red, Green or Blue LEDs or lasers).

I²C should be connected to the host processor for sending commands to the DLPC3479. The only power supplies needed external to the projector are the battery (SYSPWR) and a regulated 1.8-V supply. A single signal (PROJ_ON) controls the entire DLP system power. When PROJ_ON is high, the DLP system turns on and when PROJ_ON is low, the DLPC3479 turns off and draws only a few microamperes of current on SYSPWR. When PROJ_ON is low, the 1.8-V power supply can remain at 1.8 V for use by other sub systems. When PROJ_ON is low, the DLPA3000 draws no current on the 1.8-V supply.

The TSTPT_2 pin on the master controller outputs a 25ns pulse width that should be connected to the 3DR (input) pin of the slave controller. In case VCC_INTF is not set to 1.8V, a voltage translator is required. The propagation delay between the rising edge of TSTPT_2 pin on the master controller and the VIH of 3DR (input) pin on slave controller is recommended to be under 10ns.

9.2.2 Detailed Design Procedure

For connecting together the DLP4710 (.47 1080p) DMD, the 2xDLPC3479 controller, and the DLPA3000 or DLPA3005 device PMIC/LED driver, see the reference design schematic. When a circuit board layout is created from this schematic, a very small circuit board is possible. An example small board layout is included in the reference design data base. Follow the layout guidelines to achieve a reliable system.

9.2.3 Application Curve

As the LED currents that are driven time-sequentially through the red, green, and blue LEDs are increased, the brightness of the projector increases. This increase is somewhat non-linear, and the curve for typical white screen lumens changes with LED currents is shown in 图 32 when using the DLPA3000 or DLPA3005 device . For the LED currents shown, it is assumed that the same current amplitude is applied to the red, green, and blue LEDs.

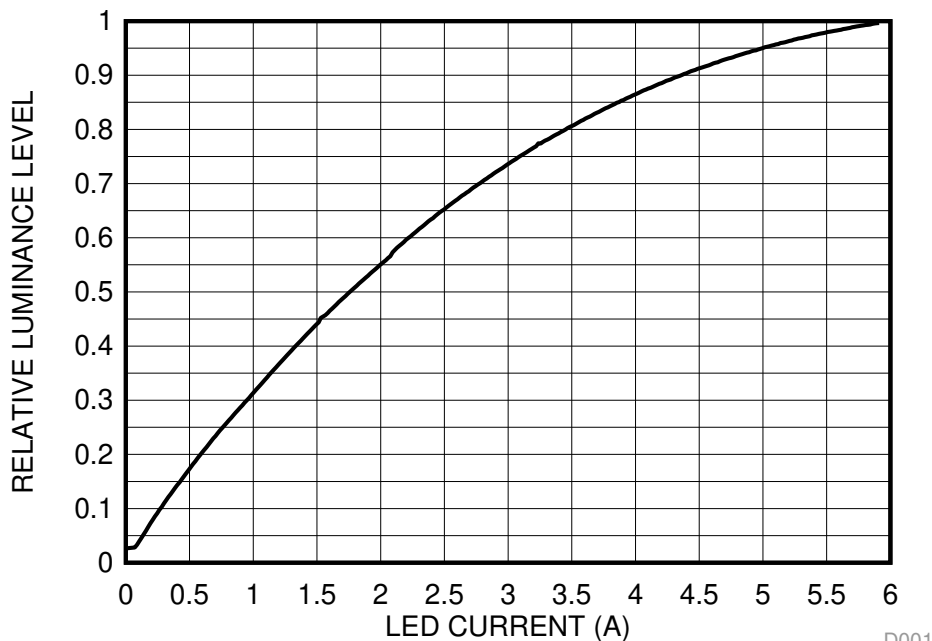


图 32. Luminance vs Current

D001

10 Power Supply Recommendations

10.1 DLPC3479 System Design Consideration

System power regulation: It is acceptable for $V_{DD_PLL D}$ and $V_{DD_PLL M}$ to be derived from the same regulator as the core V_{DD} , but to minimize the AC noise component they should be filtered as recommended in the [PCB Layout Guidelines for Internal ASIC PLL Power](#).

10.2 System Power-Up and Power-Down Sequence

Although the DLPC3479 requires an array of power supply voltages, (for example, V_{DD} , $V_{DDL P12}$, $V_{DD_PLL M/D}$, V_{CC18} , V_{CC_FLSH} , V_{CC_INTF}), if $V_{DDL P12}$ is tied to the 1.1-V V_{DD} supply (which is assumed to be the typical configuration), then there are no restrictions regarding the relative order of power supply sequencing to avoid damaging the DLPC3479. (This is true for both power-up and power-down scenarios). Similarly, there is no minimum time between powering-up or powering-down the different supplies if $V_{DDL P12}$ is tied to the 1.1-V V_{DD} supply.

If however $V_{DDL P12}$ is not tied to the V_{DD} supply, then $V_{DDL P12}$ must be powered-on after the V_{DD} supply is powered-on, and powered-off before the V_{DD} supply is powered-off. In addition, if $V_{DDL P12}$ is not tied to V_{DD} , then $V_{DDL P12}$ and V_{DD} supplies should be powered on or powered off within 100 ms of each other.

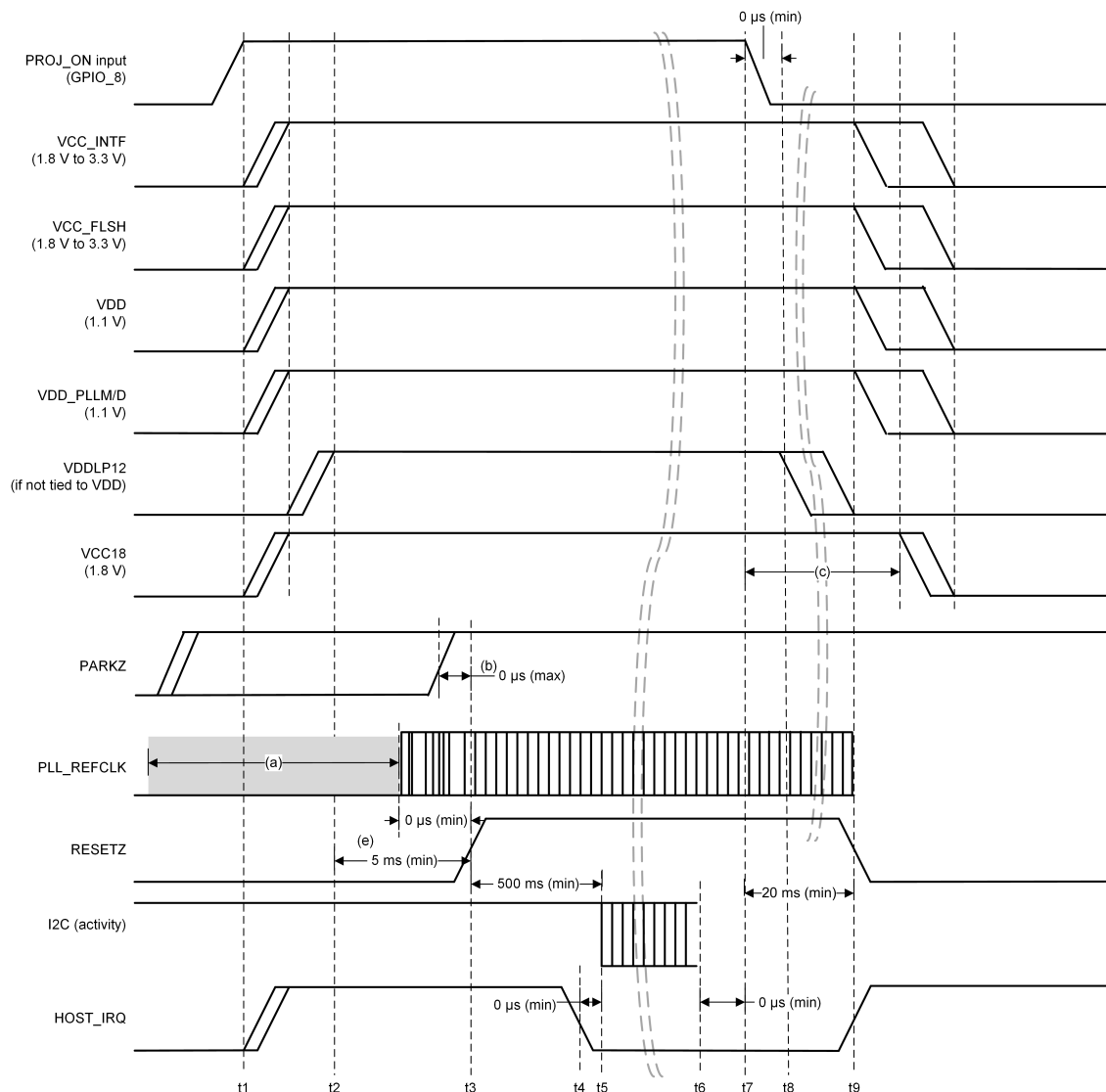
Although there is no risk of damaging the DLPC3479 if the above power sequencing rules are followed, the following additional power sequencing recommendations must be considered to ensure proper system operation.

- To ensure that DLPC3479 output signal states behave as expected, all DLPC3479 I/O supplies should remain applied while V_{DD} core power is applied. If V_{DD} core power is removed while the I/O supply (V_{CC_INTF}) is applied, then the output signal state associated with the inactive I/O supply will go to a high impedance state.
- Additional power sequencing rules may exist for devices that share the supplies with the DLPC3479, and thus these devices may force additional system power sequencing requirements.

Note that when V_{DD} core power is applied, but I/O power is not applied, additional leakage current may be drawn. This added leakage does not affect normal DLPC3479 operation or reliability.

[Figure 33](#) and [Figure 34](#) show the DLPC3479 power-up and power-down sequence for both the normal PARK and fast PARK operations of the DLPC3479 ASIC.

System Power-Up and Power-Down Sequence (接下页)



t1: HOST_IRQ is driven high when power and RESETZ are applied to indicate the DLPC3479 is not ready for operation, and then is driven low after initialization is complete.

t2: Point at which all supplies reach 95% of their specified nominal values.

t4: I2C access can start immediately after HOST_IRQ goes low (this should occur within 500 ms from the release of RESETZ).

t6: I2C activity should cease immediately upon de-assertion on PROJ_ON.

t8: Point at which all supplies reach 95% of their specified nominal values.

(a) PLL_REFCLK may be active before power is applied.

(b) PARKZ must be set high a minimum of 0 μ s before RESETZ is released to support auto-initialization.

(c) VCC18 must remain ON long enough to satisfy DMD power sequencing requirements defined in the DLPA200x specification.

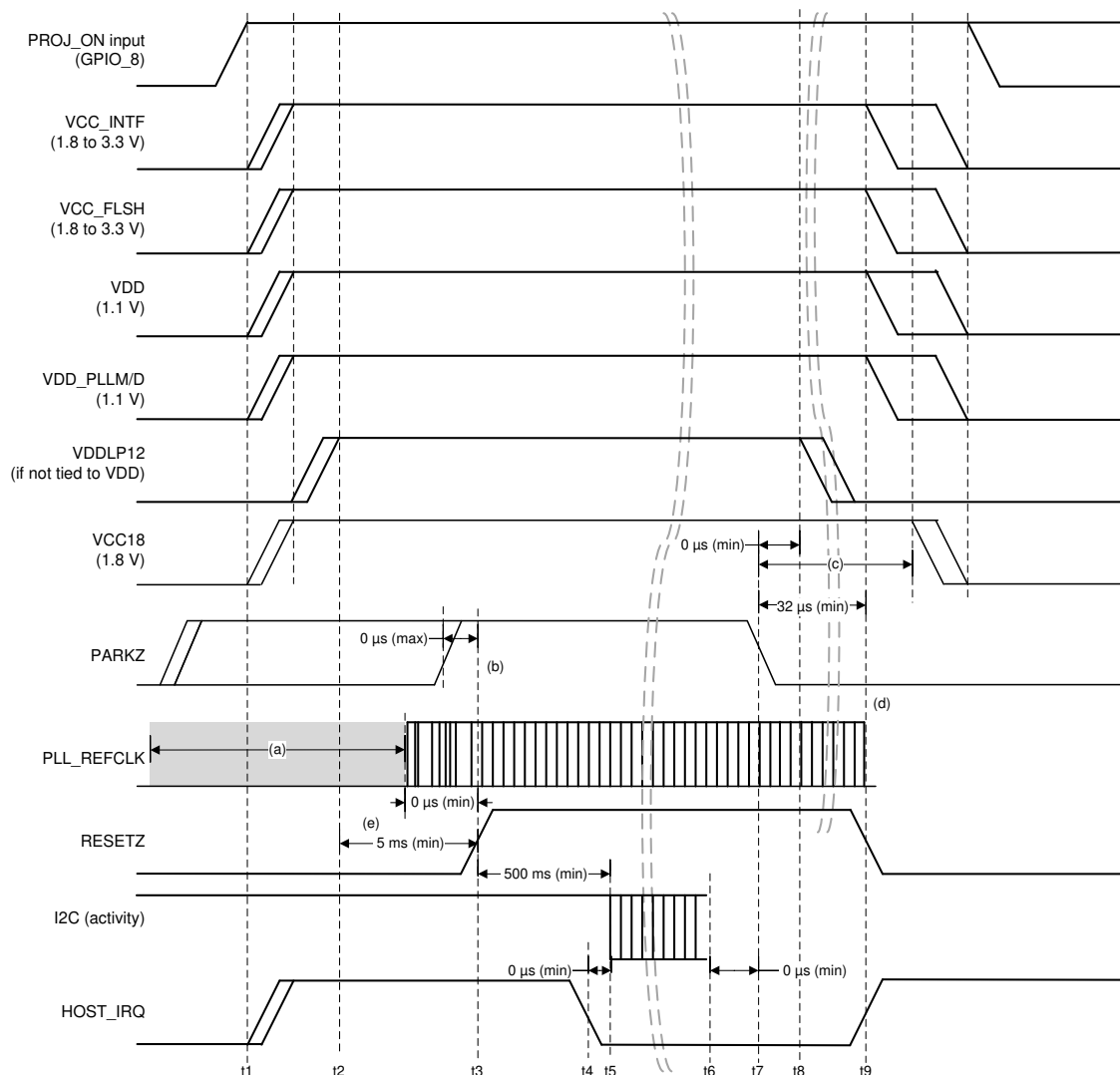
(d) PLL_REFCLK must become stable within 5 ms of all power being applied (for external oscillator application this is oscillator dependent and for crystal applications this is crystal and ASIC oscillator cell dependent).

t8: PLL_REFCLK and all ASIC supplies (except VDDL12) must remain active for a minimum of 20 ms after PROJ_ON goes low.

t9: HOST_IRQ is pulled high immediately after RESETZ is asserted low.

图 33. DLPC3479 Power-Up/PROJ_ON = 0 Initiated Normal PARK and Power-Down

System Power-Up and Power-Down Sequence (接下页)



t1: HOST_IRQ is driven high when power and RESETZ are applied to indicate the controller is not ready for operation, and then is driven low after initialization is complete

t2: all supplies reach 95% of their specified nominal values.

t4: I²C access can start immediately after HOST_IRQ goes low (this should occur within 500 ms from the release of RESETZ)

t6: I²C activity must cease immediately upon active low assertion of PARKZ T9 HOST_IRQ is pulled high immediately after RESETZ is asserted low.

t8: All power supplies reach 95% of their specified nominal values.

t9: HOST_IRQ is pulled high immediately after RESETZ is asserted low.

(a) PLL_REFCLK may be active before power is applied.

(b) PARKZ must be set high a minimum of 0 μs before RESETZ is released to support auto-initialization.

©) VCC18 must remain ON long enough to satisfy DMD power sequencing requirements defined in the DLPA2000 specification.

(d) PARKZ must be set low a minimum of 32 μs before any power is removed (except VDDL12), before PLL_REFCLK is stopped and before RESETZ is asserted low to allow time for the DMD mirrors to be parked.

(e) PLL_REFCLK must become stable within 5 ms of all power being applied (for external oscillator application this is oscillator dependent and for crystal applications this is crystal and ASIC oscillator cell dependent).

图 34. DLPC3479 Power-Up/PARKZ = 0 Initiated Fast PARK and Power-Down

10.3 DLPC3479 Power-Up Initialization Sequence

It is assumed that an external power monitor will hold the DLPC3479 in system reset during power-up. It must do this by driving RESETZ to a logic low state. It should continue to assert system reset until all ASIC voltages have reached minimum specified voltage levels, PARKZ is asserted high, and input clocks are stable. During this time, most ASIC outputs will be driven to an inactive state and all bidirectional signals will be configured as inputs to avoid contention. ASIC outputs that are not driven to an inactive state are tri-stated. These include LED_SEL_0, LED_SEL_1, SPICLK, SPIDOUT, and SPICSZ0 (see RESETZ pin description for full signal descriptions in [Pin Configuration and Functions](#)). After power is stable and the PLL_REFCLK_I clock input to the DLPC3479 is stable, then RESETZ should be deactivated (set to a logic high). The DLPC3479 then performs a power-up initialization routine that first locks its PLL followed by loading self configuration data from the external flash. Upon release of RESETZ, all DLPC3479 I/Os will become active. Immediately following the release of RESETZ, the HOST_IRQ signal will be driven high to indicate that the auto initialization routine is in progress. However, since a pullup resistor is connected to signal HOST_IRQ, this signal will have already gone high before the DLPC3479 actively drives it high. Upon completion of the auto-initialization routine, the DLPC3479 will drive HOST_IRQ low to indicate the initialization done state of the DLPC3479 has been reached.

Note that the host processor can start sending I²C commands after HOST_IRQ goes low.

10.4 DMD Fast PARK Control (PARKZ)

The PARKZ signal is defined to be an early warning signal that should alert the ASIC 32 μ s before DC supply voltages have dropped below specifications in fast PARK operation. This allows the ASIC time to park the DMD, ensuring the integrity of future operation. Note that the reference clock should continue to run and RESETZ should remain deactivated for at least 32 μ s after PARKZ has been deactivated (set to a logic low) to allow the park operation to complete.

10.5 Hot Plug Usage

The DLPC3479 provides fail-safe I/O on all host interface signals (signals powered by V_{CC_INTF}). This allows these inputs to be driven high even when no I/O power is applied. Under this condition, the DLPC3479 will not load the input signal nor draw excessive current that could degrade ASIC reliability. For example, the I²C bus from the host to other components would not be affected by powering off V_{CC_INTF} to the DLPC3479. TI recommends weak pullups or pulldowns on signals feeding back to the host to avoid floating inputs.

If the I/O supply (V_{CC_INTF}) is powered off, but the core supply (V_{DD}) is powered on, then the corresponding input buffer may experience added leakage current, but this does not damage the DLPC3479.

10.6 Maximum Signal Transition Time

Unless otherwise noted, 10 ns is the maximum recommended 20 to 80% rise or fall time to avoid input buffer oscillation. This applies to all DLPC3479 input signals. However, the PARKZ input signal includes an additional small digital filter that ignores any input buffer transitions caused by a slower rise or fall time for up to 150 ns.

11 Layout

11.1 Layout Guidelines

11.1.1 PCB Layout Guidelines for Internal ASIC PLL Power

The following guidelines are recommended to achieve desired ASIC performance relative to the internal PLL. Each DLPC3479 contains 2 internal PLLs which have dedicated analog supplies (V_{DD_PLLX} , V_{SS_PLLX} , V_{DD_PLLD} , V_{SS_PLLD}). As a minimum, V_{DD_PLLX} power and V_{SS_PLLX} ground pins should be isolated using a simple passive filter consisting of two series Ferrites and two shunt capacitors (to widen the spectrum of noise absorption). It's recommended that one capacitor be a 0.1- μ F capacitor and the other be a 0.01- μ F capacitor. All four components should be placed as close to the ASIC as possible but it's especially important to keep the leads of the high frequency capacitors as short as possible. Note that both capacitors should be connected across V_{DD_PLLX} and V_{SS_PLLX} / V_{DD_PLLD} and V_{SS_PLLD} respectfully on the ASIC side of the Ferrites.

For the ferrite beads used, their respective characteristics should be as follows:

- DC resistance less than 0.40 Ω
- Impedance at 10 MHz equal to or greater than 180 Ω
- Impedance at 100 MHz equal to or greater than 600 Ω

The PCB layout is critical to PLL performance. It is vital that the quiet ground and power are treated like analog signals. Therefore, V_{DD_PLLX} and V_{DD_PLLD} must be a single trace from each DLPC3479 to both capacitors and then through the series ferrites to the power source. The power and ground traces should be as short as possible, parallel to each other, and as close as possible to each other.

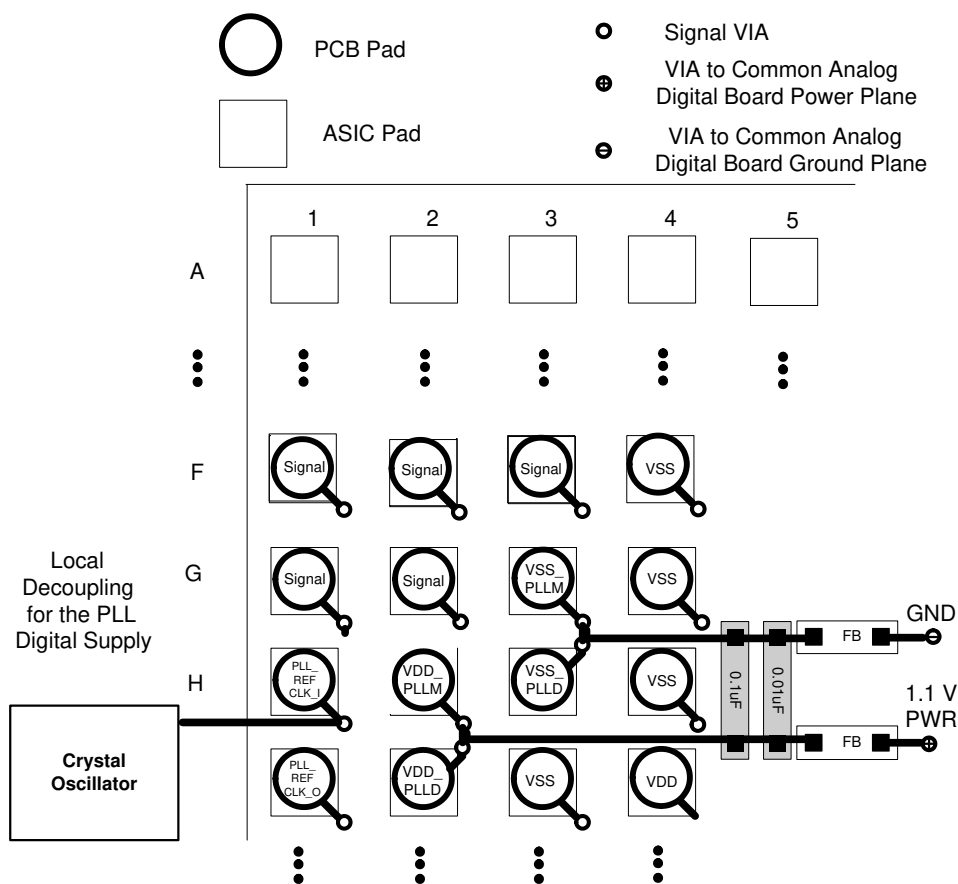


图 35. PLL Filter Layout

Layout Guidelines (接下页)

11.1.2 DLPC3479 Reference Clock

The DLPC3479 requires an external reference clock to feed its internal PLL. A crystal oscillator can supply this reference. For flexibility, the DLPC3479 accepts either of two reference clock frequencies, but both must have a maximum frequency variation of ± 200 ppm (including aging, temperature, and trim component variation).

The two DLPC3479 devices require a single dedicated oscillator where the oscillator output drives both DLPC3479 devices. The oscillator must drive the PLL_REFCLK_I pin on each DLPC3479. Leave the PLL_REFCLK_O pins unconnected.

The external oscillator must be able to drive at least a 15-pF load. Routing length from the oscillator to each DLPC3479 should be closely matched.

11.1.3 General PCB Recommendations

TI recommends 1-oz. copper planes in the PCB design to achieve needed thermal connectivity.

11.1.4 General Handling Guidelines for Unused CMOS-Type Pins

To avoid potentially damaging current caused by floating CMOS input-only pins, TI recommends that unused ASIC input pins be tied through a pullup resistor to its associated power supply or a pulldown to ground. For ASIC inputs with an internal pullup or pulldown resistors, it is unnecessary to add an external pullup or pulldown unless specifically recommended. Note that internal pullup and pulldown resistors are weak and should not be expected to drive the external line. The DLPC3479 implements very few internal resistors and these are noted in the pin list. When external pullup or pulldown resistors are needed for pins that have built-in weak pullups or pulldowns, use the value 8 k Ω (max).

Unused output-only pins should never be tied directly to power or ground, but can be left open.

When possible, TI recommends that unused bidirectional I/O pins be configured to their output state such that the pin can be left open. If this control is not available and the pins may become an input, then they should be pulled-up (or pulled-down) using an appropriate, dedicated resistor.

Layout Guidelines (接下页)

11.1.5 Maximum Pin-to-Pin, PCB Interconnects Etch Lengths

表 12. Max Pin-to-Pin PCB Interconnect Recommendations⁽¹⁾⁽²⁾

DMD BUS SIGNAL	SIGNAL INTERCONNECT TOPOLOGY		UNIT
	SINGLE BOARD SIGNAL ROUTING LENGTH	MULTI-BOARD SIGNAL ROUTING LENGTH	
DMD_HS_CLK_P DMD_HS_CLK_N	6.0 152.4	See ⁽³⁾	inch (mm)
DMD_HS_WDATA_A_P DMD_HS_WDATA_A_N	6.0 152.4	See ⁽³⁾	inch (mm)
DMD_HS_WDATA_B_P DMD_HS_WDATA_B_N			
DMD_HS_WDATA_C_P DMD_HS_WDATA_C_N			
DMD_HS_WDATA_D_P DMD_HS_WDATA_D_N			
DMD_HS_WDATA_E_P DMD_HS_WDATA_E_N			
DMD_HS_WDATA_F_P DMD_HS_WDATA_F_N			
DMD_HS_WDATA_G_P DMD_HS_WDATA_G_N			
DMD_HS_WDATA_H_P DMD_HS_WDATA_H_N			
DMD_LS_CLK	6.5 165.1	See ⁽³⁾	inch (mm)
DMD_LS_WDATA	6.5 165.1	See ⁽³⁾	inch (mm)
DMD_LS_RDATA	6.5 165.1	See ⁽³⁾	inch (mm)
DMD_DEN_ARSTZ	7.0 177.8	See ⁽³⁾	inch (mm)

(1) Maximum signal routing length includes escape routing.

(2) Multi-board DMD routing length is more restricted due to the impact of the connector.

(3) Due to board variations, these are impossible to define. Any board designs should SPICE simulate with the ASIC IBIS models to ensure single routing lengths do not exceed requirements.

表 13. High Speed PCB Signal Routing Matching Requirements⁽¹⁾⁽²⁾⁽³⁾⁽⁴⁾

SIGNAL GROUP LENGTH MATCHING				
INTERFACE	SIGNAL GROUP	REFERENCE SIGNAL	MAX MISMATCH ⁽⁵⁾	UNIT
DMD	DMD_HS_WDATA_A_P DMD_HS_WDATA_A_N	DMD_HS_CLK_P DMD_HS_CLK_N	±1.0 (±25.4)	inch (mm)
	DMD_HS_WDATA_B_P DMD_HS_WDATA_B_N			
	DMD_HS_WDATA_C_P DMD_HS_WDATA_C_N			
	DMD_HS_WDATA_D_P DMD_HS_WDATA_D_N			
	DMD_HS_WDATA_E_P DMD_HS_WDATA_E_N			
	DMD_HS_WDATA_F_P DMD_HS_WDATA_F_N			
	DMD_HS_WDATA_G_P DMD_HS_WDATA_G_N			
	DMD_HS_WDATA_H_P DMD_HS_WDATA_H_N			
DMD	DMD_HS_WDATA_x_P	DMD_HS_WDATA_x_N	±0.025 (±0.635)	inch (mm)
	DMD_HS_CLK_P	DMD_HS_CLK_N	±0.025 (±0.635)	inch (mm)
DMD	DMD_LS_WDATA DMD_LS_RDATA	DMD_LS_CLK	±0.2 (±5.08)	inch (mm)
DMD	DMD_DEN_ARSTZ	N/A	N/A	inch (mm)

- (1) These values apply to PCB routing only. They do not include any internal package routing mismatch associated with the DLPC3479 or the DMD.
- (2) DMD HS data lines are differential, thus these specifications are pair-to-pair.
- (3) Training is applied to DMD HS data lines, so defined matching requirements are slightly relaxed.
- (4) DMD LS signals are single ended.
- (5) Mismatch variance for a signal group is always with respect to reference signal.

11.1.6 Number of Layer Changes

- Single-ended signals: Minimize the number of layer changes.
- Differential signals: Individual differential pairs can be routed on different layers, but the signals of a given pair should not change layers.

11.1.7 Stubs

- Stubs should be avoided.

11.1.8 Terminations

- No external termination resistors are required on DMD_HS differential signals.
- The DMD_LS_CLK and DMD_LS_WDATA signal paths should include a 43-Ω series termination resistor located as close as possible to the corresponding ASIC pins.
- The DMD_LS_RDATA signal path should include a 43-Ω series termination resistor located as close as possible to the corresponding DMD pin.
- DMD_DEN_ARSTZ does not require a series resistor.

11.1.9 Routing Vias

- The number of vias on DMD_HS signals should be minimized and should not exceed two.
- Any and all vias on DMD_HS signals should be located as close to the ASIC as possible.
- The number of vias on the DMD_LS_CLK and DMD_LS_WDATA signals should be minimized and not exceed two.
- Any and all vias on the DMD_LS_CLK and DMD_LS_WDATA signals should be located as close to the ASIC

as possible.

11.2 Layout Example

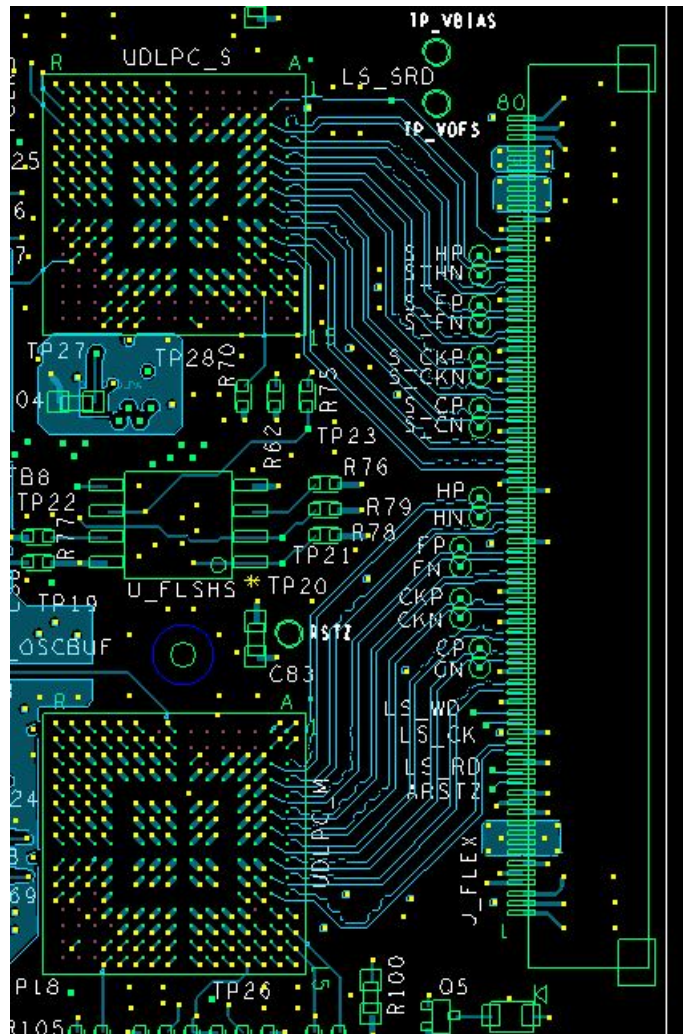


图 36. Board Layout

11.3 Thermal Considerations

The underlying thermal limitation for the DLPC3479 is that the maximum operating junction temperature (T_J) cannot be exceeded (as specified in [Recommended Operating Conditions](#) table). This temperature depends on operating ambient temperature, airflow, PCB design (including the component layout density and the amount of copper used), power dissipation of the DLPC3479, and power dissipation of surrounding components. The DLPC3479 package extracts heat through the power and ground planes of the PCB. Make sure to consider copper content and airflow over the PCB when confirming temperature specifications of the application.

The recommended maximum operating ambient temperature (T_A) is provided primarily as a design target. This target is based on the maximum DLPC3479 device power dissipation and $R_{\theta JA}$ at 0 ms of forced airflow, where $R_{\theta JA}$ is the thermal resistance of the package as measured using a glater test PCB with two, 1-oz power planes. This JEDEC test PCB is not necessarily representative of the DLPC3479 PCB. The reported thermal resistance may not be accurate in the actual product application. Although the actual thermal resistance may be different, it is the best information available during the design phase to estimate thermal performance. However, after the PCB is designed and the product is built, TI highly recommends that thermal performance is measured and validated.

Thermal Considerations (接下页)

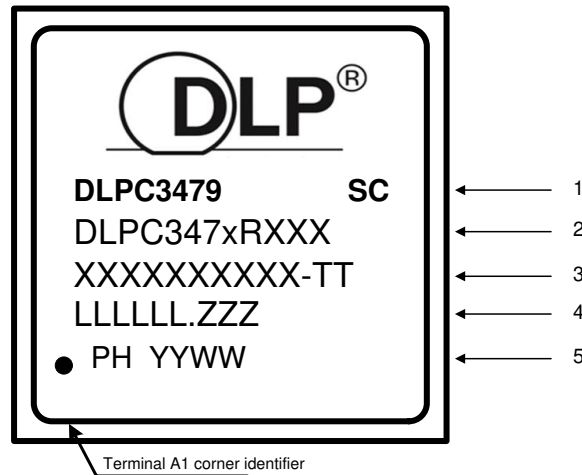
Measure the top center case temperature under the worse case product scenario (maximum power dissipation, maximum voltage, maximum ambient temperature) and validated not to exceed the maximum recommended case temperature (T_C). This specification is based on the measured ϕ_{JT} for the DLPC3479 package and provides a relatively accurate correlation to junction temperature. Take care when measuring this case temperature to prevent accidental cooling of the package surface. TI recommends a small (approximately 40 gauge) thermocouple. The bead and thermocouple wire should contact the top of the package and be covered with a minimal amount of thermally conductive epoxy. The wires should be routed closely along the package and the board surface to avoid cooling the bead through the wires.

12 器件和文档支持

12.1 器件支持

12.1.1 器件命名规则

12.1.1.1 器件标记



标记定义:

- 第 1 行: DLP® 器件名称: DLPC347x, 其中 **x** 表示器件名称 ID, 此处为 9。
SC: 焊球成分
e1: 表示含有 SnAgCu 的无铅焊球。
G8: 表示含有锡银铜 (SnAgCu) 的无铅焊球, 其中银含量小于或等于 1.5% 且模压混合物符合 TI 的“绿色环保”定义。
- 第 2 行: TI 器件型号
DLP® 器件名称: DLPC347x, 其中 **x** 表示器件名称 ID, 此处为 9。
R 表示 TI 器件版次代字, 例如 A、B 或 C。
XXX 表示器件封装标识符。
- 第 3 行: XXXXXXXXXXXX-TT 制造商部件号
- 第 4 行: LLLLLL.ZZZ 半导体晶圆的铸造批次代码
LLLLLL: 生产批次代码
ZZZ: 分批编号
- 第 5 行: XX YYWW ES: 封装组件信息
XX: 制造工厂
YYWW: 日期代码 (YY = 年, WW = 周)

注

1. 工程原型样品则在 TI 部件号之后加 **X** 后缀表示。例如: 2512737-0001X。
2. 有关 DLPC3479 可在 DMD 上支持的分辨率信息, 请按照器件编号, 参阅 [Input Source - Frame Rates](#)。

12.1.2 视频时序参数定义

每帧有效扫描行数 (**ALPF**) 定义一帧中包含可显示数据的行数: ALPF 是每帧总行数 (TLPF) 的子集。

每行有效像素 (**APPL**) 定义包含可显示数据的一行中的像素时钟数: APPL 是每行总像素 (TPPL) 的子集。

水平后沿 (**HBP**) 消隐 水平同步之后, 第一个有效像素之前的消隐像素时钟数量。注意: HBP 时间以各自同步信号

器件支持 (接下页)

的前沿（有效）为基准。

水平前沿 (HFP) 消隐 最后一个有效时钟之后，水平同步之前的消隐像素时钟的数量。

水平同步 (HS) 定义水平间隔（行）开始的时序基准点。绝对基准点由 HS 信号的有效边沿定义。有效边沿（源定义的上升沿或下降沿）是测量所有水平消隐参数的基准。

每帧总行数 (TLPF) 以行数定义垂直扫描时间（帧时间）： $TLPF = \text{每帧总行数（有效和无效行）}$ 。

每行总像素 (TPPL) 以像素时钟数定义水平扫描时间： $TPPL = \text{每行总像素时钟数（有效和无效像素时钟）}$ 。

垂直同步 (VS) 定义垂直间隔（帧）开始的时序基准点。这个绝对基准点由 VS 信号的有效边沿定义。有效边沿（源定义的上升沿或下降沿）是测量所有垂直消隐参数的基准。

垂直后沿 (VBP) 消隐 垂直同步的前沿之后，第一个有效行之前的消隐行的数量。

垂直前沿 (VFP) 消隐 最后一个有效行之后，垂直同步的前沿之前的消隐行的数量。

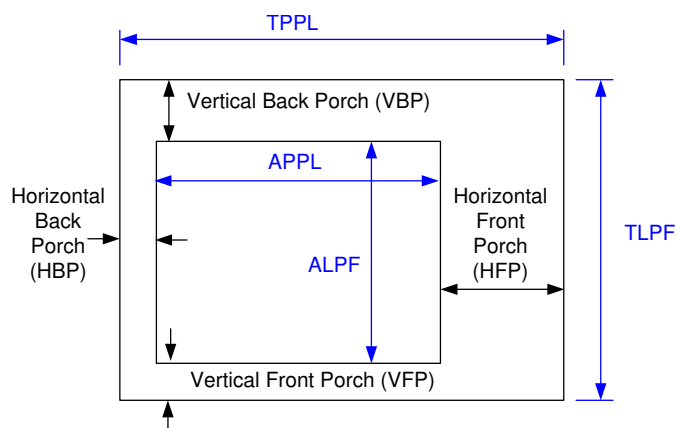


图 37. 时序参数图

12.2 文档支持

下表列出了快速访问链接。类别包括技术文档、支持与社区资源、工具和软件，以及申请样片或购买产品的快速链接。

表 14. 相关文档

器件	产品文件夹	样片与购买	技术文档	工具与软件	支持和社区
DLP4710	单击此处	单击此处	单击此处	单击此处	单击此处
DLPA3005	单击此处	单击此处	单击此处	单击此处	单击此处
DLPA3000	请单击此处	单击此处	单击此处	单击此处	单击此处

12.3 接收文档更新通知

要接收文档更新通知，请导航至 TI.com.cn 上的器件产品文件夹。单击右上角的 [通知我](#) 进行注册，即可每周接收产品信息更改摘要。有关更改的详细信息，请查看任何已修订文档中包含的修订历史记录。

12.4 社区资源

The following links connect to TI community resources. Linked contents are provided "AS IS" by the respective contributors. They do not constitute TI specifications and do not necessarily reflect TI's views; see TI's [Terms of Use](#).

TI E2E™ Online Community *TI's Engineer-to-Engineer (E2E) Community*. Created to foster collaboration among engineers. At e2e.ti.com, you can ask questions, share knowledge, explore ideas and help solve problems with fellow engineers.

Design Support *TI's Design Support* Quickly find helpful E2E forums along with design support tools and contact information for technical support.

12.5 商标

IntelliBright, E2E are trademarks of Texas Instruments.

12.6 静电放电警告



这些装置包含有限的内置 ESD 保护。存储或装卸时，应将导线一起截短或将装置放置于导电泡棉中，以防止 MOS 门极遭受静电损伤。

12.7 Glossary

[SLYZ022](#) — *TI Glossary*.

This glossary lists and explains terms, acronyms, and definitions.

13 机械、封装和可订购信息

以下页面包含机械、封装和可订购信息。这些信息是指定器件的最新可用数据。数据如有变更，恕不另行通知，且不会对此文档进行修订。如需获取此数据表的浏览器版本，请查阅左侧的导航栏。

PACKAGING INFORMATION

Orderable Device	Status (1)	Package Type	Package Drawing	Pins	Package Qty	Eco Plan (2)	Lead finish/ Ball material (6)	MSL Peak Temp (3)	Op Temp (°C)	Device Marking (4/5)	Samples
DLPC3479CZEZ	ACTIVE	NFBGA	ZEZ	201	119	Non-RoHS & Non-Green	Call TI	Call TI		(DLPC3479 G8, DLP C3479 G8) DLPC3479CZEZ ECP292548C-9G	

⁽¹⁾ The marketing status values are defined as follows:

ACTIVE: Product device recommended for new designs.

LIFEBUY: TI has announced that the device will be discontinued, and a lifetime-buy period is in effect.

NRND: Not recommended for new designs. Device is in production to support existing customers, but TI does not recommend using this part in a new design.

PREVIEW: Device has been announced but is not in production. Samples may or may not be available.

OBSOLETE: TI has discontinued the production of the device.

⁽²⁾ **RoHS:** TI defines "RoHS" to mean semiconductor products that are compliant with the current EU RoHS requirements for all 10 RoHS substances, including the requirement that RoHS substance do not exceed 0.1% by weight in homogeneous materials. Where designed to be soldered at high temperatures, "RoHS" products are suitable for use in specified lead-free processes. TI may reference these types of products as "Pb-Free".

RoHS Exempt: TI defines "RoHS Exempt" to mean products that contain lead but are compliant with EU RoHS pursuant to a specific EU RoHS exemption.

Green: TI defines "Green" to mean the content of Chlorine (Cl) and Bromine (Br) based flame retardants meet JS709B low halogen requirements of <=1000ppm threshold. Antimony trioxide based flame retardants must also meet the <=1000ppm threshold requirement.

⁽³⁾ MSL, Peak Temp. - The Moisture Sensitivity Level rating according to the JEDEC industry standard classifications, and peak solder temperature.

⁽⁴⁾ There may be additional marking, which relates to the logo, the lot trace code information, or the environmental category on the device.

⁽⁵⁾ Multiple Device Markings will be inside parentheses. Only one Device Marking contained in parentheses and separated by a "~" will appear on a device. If a line is indented then it is a continuation of the previous line and the two combined represent the entire Device Marking for that device.

⁽⁶⁾ Lead finish/Ball material - Orderable Devices may have multiple material finish options. Finish options are separated by a vertical ruled line. Lead finish/Ball material values may wrap to two lines if the finish value exceeds the maximum column width.

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NFBGA - 1 mm max height

Figure 1: Dimensions and details of the test specimen.

The figure shows three views of the test specimen:

- Top View:** A square specimen with overall dimensions of 13.1 x 13.1 (12.9 x 12.9). A shaded corner is labeled "BALL A1 CORNER".
- Side View:** A cross-section showing the specimen's thickness. The maximum thickness is 1. The typical thickness is 0.31/0.21. The "BALL TYP" layer is shown. A "SEATING PLANE" is indicated with a 0.1 C tolerance.
- Detail View:** A 15x15 grid of balls (201X Ø 0.4 / 0.3) with a 11.2 TYP pitch and 0.8 TYP edge distance. The grid is labeled with letters A-R and numbers 1-15. A "SYMM" (Symmetry) line is shown.

The table below provides the tolerance values for various features:

Feature	Symbol	Value	Material	Condition	Feature	Symbol	Value	Material	Condition
Ball Diameter	Ø	0.15	M	C	Ball Diameter	Ø	0.08	M	C
Ball Position	⊕	0.15	M	C	Ball Position	⊕	0.08	M	C
Ball Position	⊕	0.15	M	C	Ball Position	⊕	0.08	M	C
Ball Position	⊕	0.15	M	C	Ball Position	⊕	0.08	M	C



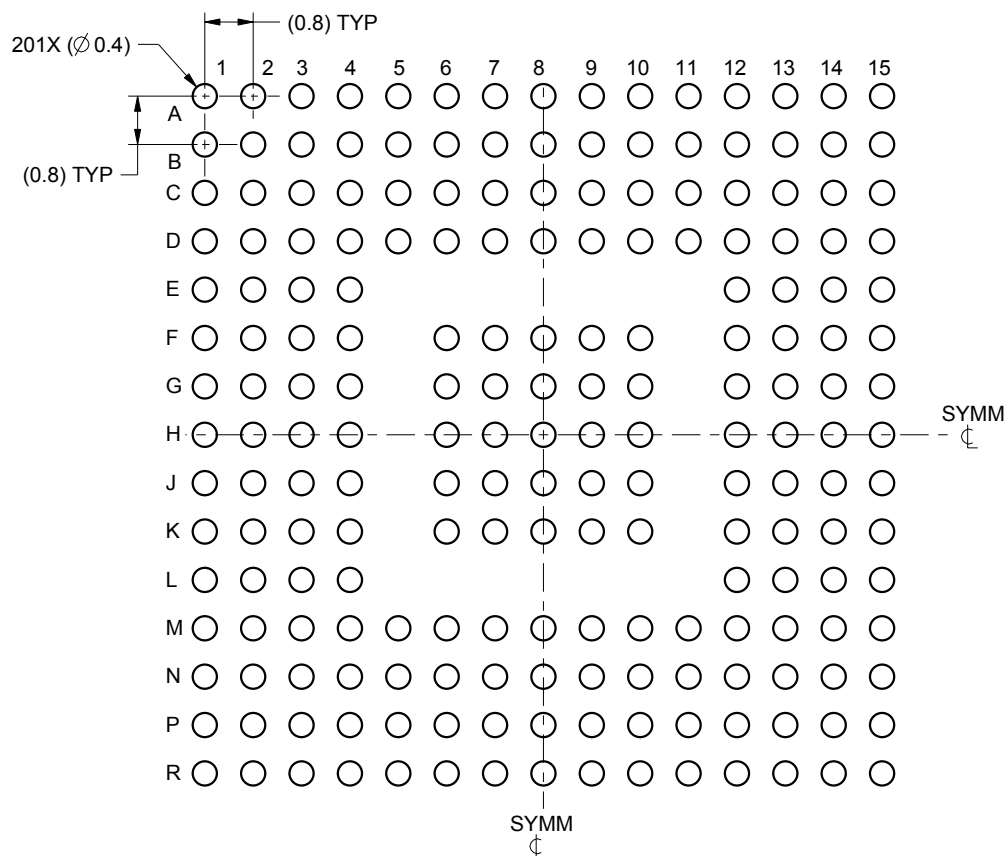
TEXAS
INSTRUMENTS
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EXAMPLE BOARD LAYOUT

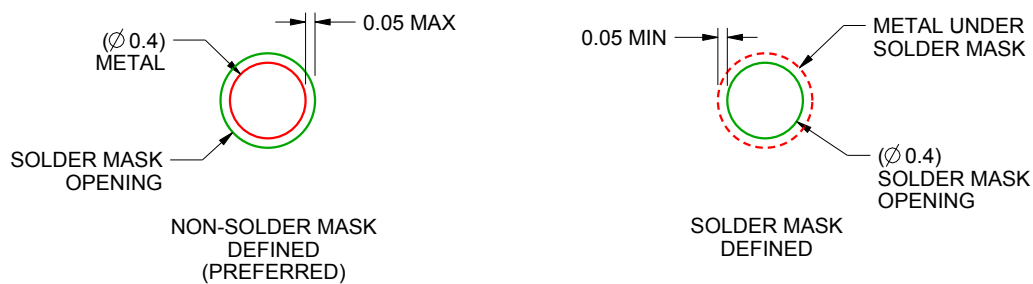
ZEZ0201A

NFBGA - 1 mm max height

PLASTIC BALL GRID ARRAY



LAND PATTERN EXAMPLE
SCALE:8X



SOLDER MASK DETAILS
NOT TO SCALE

4221521/A 03/2015

NOTES: (continued)

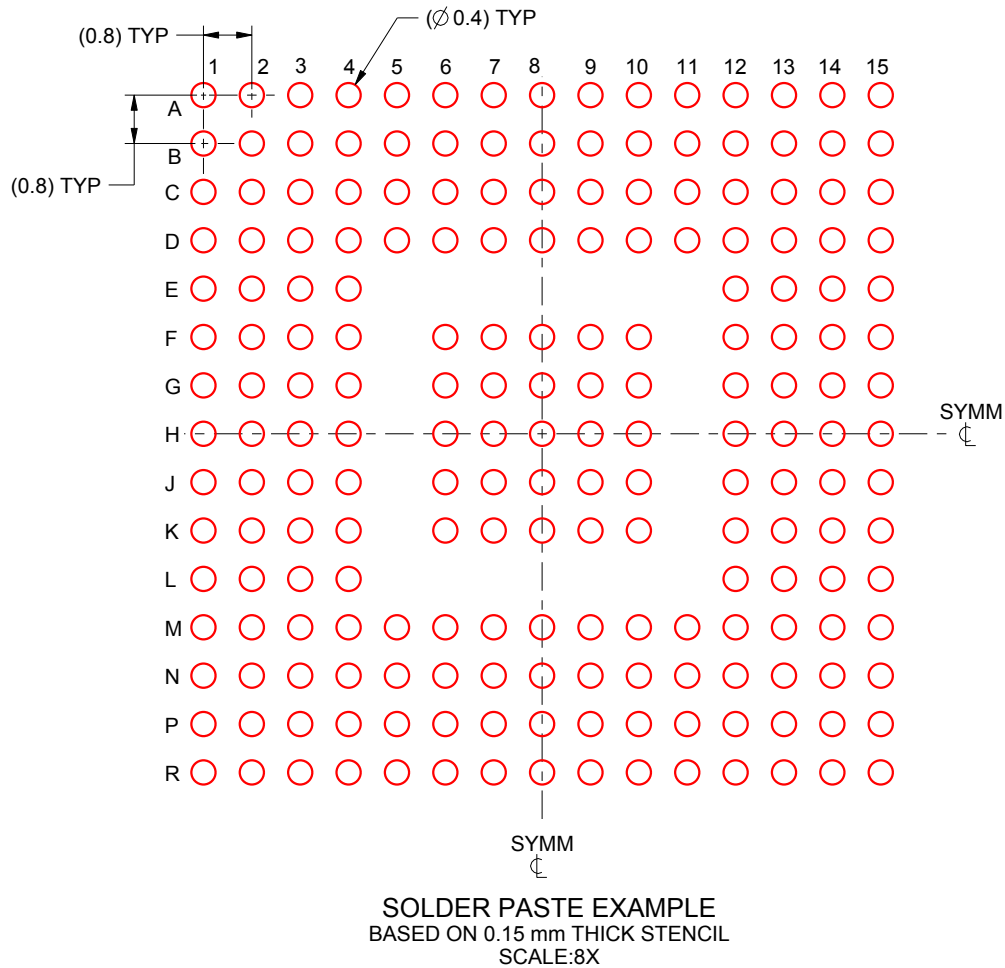
- Final dimensions may vary due to manufacturing tolerance considerations and also routing constraints. For information, see Texas Instruments literature number SPRAA99 (www.ti.com/lit/spraa99).

EXAMPLE STENCIL DESIGN

ZEZ0201A

NFBGA - 1 mm max height

PLASTIC BALL GRID ARRAY



4221521/A 03/2015

NOTES: (continued)

4. Laser cutting apertures with trapezoidal walls and rounded corners may offer better paste release.

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